

Regulatory Limits to Risk Management

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Variable annuities, the largest liability of U.S. life insurers, are investment products containing long-dated minimum return guarantees. I show that guarantees with similar economic risks are treated differently by regulation and these differences impact insurers' hedging behavior. When the regulatory regime recognizes certain risks, insurers start to hedge these risks in a substantial way. For some guarantees, this involves hedging both interest rate and equity market risks. However, for others, it involves hedging only equity market risk. As the regulatory regime still does not recognize the interest rate risk of all guarantees, insurers remain exposed to substantial interest rate risk. (*JEL* G22, G23, G28, G32, G11)

Received September 13, 2019; editorial decision November 10, 2021 by Editor Stijn Van Nieuwerburgh. Authors have furnished an Internet Appendix, which is available on the Oxford University Press Web site next to the link to the final published paper online.

Variable annuities (VAs) are the largest liability of life insurers in the United States, with over \$2 trillion outstanding as of 2018. These products, which combine mutual funds with long-dated minimum return guarantees, fulfill a growing demand for stable retirement income at a time when state pensions and defined-benefit plans have declined in importance. VAs are also a key source of risk for insurers, as the value of the guarantees provided to beneficiaries can rise substantially when equity markets fall or interest rates decline. For example,

I am deeply grateful to my advisor, Ralph Koijen, for his generous and invaluable guidance, to my committee members Anna Pavlova, Helene Rey, and Stephen Schaefer for their valuable feedback, and to Adi Sunderam, Sam Hanson, and David Scharfstein for their generous help during the paper's revision. I thank the editor, Stijn Van Nieuwerburgh, and two anonymous referees for their valuable suggestions that substantially improved the paper. For comments and discussions, I also thank Svetlana Bryzgalova, Joao Cocco, Francesca Cornelli, Nuno Clara, Alejandro Drexler, Darrell Duffie, Douglas Diamond, Andrew Ellul, Francisco Gomes, Robin Greenwood, Valentin Haddad, David Humphry, Novian Junus, Christian Laux, Alan Moreira, Zohair Motiwalla, Ken Mungan, Narayan Naik, Adriano Rampini, Arkodipta Sarkar, David Schoenherr, Varun Sharma, Peter Sun, Vikrant Vig, Vish Viswanathan, seminar participants at the Bank of England, Chicago Booth, Harvard Business School, INSEAD, London Business School, London School of Economics, MIT Sloan, Princeton, Simon Rochester, Stanford GSB, UCLA Anderson, Reserve Bank of India, 2020 IMF Macro-Financial Research Conference, 2020 MFA Meetings, 2019 EFA Meetings, 2019 ACPR Workshop on Systemic Risk and Regulation, 2018 EFA Doctoral Tutorial, and the 2018 Emerging Scholars Conference. I gratefully acknowledge financial support from the AQR Asset Management Institute and the Alfred P. Sloan Foundation, Becker Friedman Institute, The National Association of Insurance Commissioners, SNL Financial, A.M. Best Company, and Morningstar Annuity own the copyright to their respective data, which I used with their permission. Supplementary data can be found on *The Review of Financial Studies* web site. Correspondence: Ishita Sen, Harvard Business School, Boston, MA 02163; email: isen@hbs.edu.

The Review of Financial Studies 36 (2023) 2175–2223

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<https://doi.org/10.1093/rfs/hnac083>

Advance Access Publications October 31, 2022

many U.S. insurers took significant losses on their VA liabilities during the 2008-2009 financial crisis, which prompted them to seek government bailouts, cut insurance supply, and engage in asset fire sales (Kojien and Yogo 2022; Ellul, Jotikasthira, and Lundblad 2011).¹ Given the size of insurers' VA liabilities, the way insurers manage the risks embedded in them has significant implications for the insurance sector and the wider economy.

In this paper, I examine how the design of the regulatory framework impacts insurers' risk management practices. There are two main challenges in studying how regulation affects risk management. First, detailed data on how financial institutions hedge risks and their usage of derivatives are typically unavailable. Insurers are unique because they report extremely detailed position-level data on derivatives, which makes it possible to comprehensively examine the determinants of their hedging behavior. Second, risk management decisions can be influenced by both economic and regulatory incentives. When we observe differences in hedging decisions across institutions, products or time, it is not immediately obvious whether economic or regulatory forces are driving this variation. I exploit a setting where regulatory incentives differ for products that carry very similar economic risks. This allows me to identify the extent to which hedging outcomes are shaped by regulation.

Two features of the setting are important. First, derivatives are the primary way by which insurers manage the risks associated with the guarantees embedded in VAs. The guarantees are essentially long-dated put options with complex payoffs and are exposed to both interest rate and equity market risks. These risks are far easier to hedge using derivatives than using securities because of the complexity of the payoffs, the structure of the cash flows, and the ease of dynamic trading with derivatives (Milliman 2021).

Second, while the regulatory regime for insurers has always recognized the risks of derivatives, essentially marking them to market, the way it has treated the risks of VA guarantees has evolved over time and in different ways for the different types of guarantees. Prior to 2009, the regulatory regime, for the most part, treated all VA guarantees as though they carried no interest rate risk and limited equity market risk. Thus, insurers using derivatives to hedge their VA guarantees would reduce the volatility of their economic capital but increase the volatility of their regulatory capital. In 2009, the regulatory framework was overhauled. For some VA guarantees, which I label "Risk Sensitive" (RS), the new regime recognized exposure to both interest rate and equity risks by incorporating greater risk sensitivities in how liabilities were computed. For other guarantees, which I label "Partially Risk Sensitive" (PRS), the new regime recognized only equity, but not interest rate risk.

¹ Prominent underwriters of VAs regularly feature among the most systemically risky institutions in the United States (Acharya et al. 2017). Guaranteed insurance contracts are also associated with fragility and failure of insurance companies in the United Kingdom, Europe, and Japan.

I first show that RS and PRS guarantees have very similar underlying economic risks. Standard models of risk management (e.g., Smith and Stulz 1985; Froot, Scharfstein, and Stein 1993) would imply then that insurers should hedge both types of guarantees in a similar way. In these models, economic capital is the binding constraint and, therefore, only the volatility of economic (not regulatory) capital is hedged. If the volatility of regulatory capital is costly, insurers could operate with greater capital *ex ante* and continue hedging their economic exposures (Froot and Stein 1998). However, if raising capital is costly and regulatory capital is the binding constraint, then hedging would differ across guarantees: interest rate risk hedging would diverge over time, increasing only for RS, but not for PRS, guarantees, but equity risk hedging would increase in a similar way for all guarantees.

I measure hedging by constructing a long history of risk exposures using holdings data on derivatives. Insurers report individual derivative positions, including all instruments (swaps, options, futures, and forwards) for each risk factor being hedged. Detailed contract characteristics are available for each position, including direction, maturity, and strike. Thus, as opposed to using the gross notional amounts, which are commonly used in the literature due to data limitations, I can measure hedging more accurately by aggregating up individual positions. The data start in 2004. This allows me to study how hedging has evolved under different regulatory regimes and market conditions.

Using a difference-in-differences strategy that exploits variation in regulatory hedging incentives over time and across products, I show that insurers' hedging decisions are strongly impacted by regulatory incentives. I present two main results. First, insurers significantly increased interest rate hedging after the regulation is introduced: they hedged little before 2009, but hedging increases by more than five times by the end of the sample in 2016. Importantly, the increase in hedging is solely concentrated among insurers that are highly exposed to RS guarantees, but not to PRS guarantees. The differences in interest rate hedging for RS and PRS guarantees induced by the new regime are large. For example, for a 100-bps decline in interest rates, my estimates imply a hedging gap of 7.1 cents per dollar of capital for the average VA underwriter.² Second, in parallel results for equity hedging, I show that insurers also increased equity hedging after 2009, just like interest rate hedging. However, equity hedging increased in a similar way for both RS and PRS guarantees, such that the differences in hedging across guarantees induced by the new regime is negligible. This is consistent with their identical treatment under the new regulatory regime.

If we only observed variation in hedging over time or across coarse product groupings, it would be difficult to attribute the changes in hedging to regulation. For example, an increase in hedging over time does not by itself mean that

² To put this number in perspective, the capital of the average insurer declined by 8% during the height of the financial crisis between 2007:Q4 and 2009:Q1.

regulation impacted hedging because hedging could have increased due to other reasons that may have coincided with the regulatory shift; for example, economic exposures may have changed or the financial crisis may have changed insurers' risk appetite. Similarly, it would be difficult to conclude that regulation impacted hedging if I compared hedging across coarse product groupings as it is likely that risk exposures shift in a different manner for these products over time. The unique aspect of my setting is that I am able to compare the hedging for granular product groupings that have very similar economic exposures, but for which regulation changed differentially. In addition to variation over time and products, I also exploit variation across risk factors and find heterogeneity in which risk exposure an insurer hedges and which it does not. These choices line up with the incentives provided by regulation. Thus, making it unlikely that variation in economic incentives of hedging, for example, financial constraints or risk aversion, are driving these results.

A key question is if insurers hedge in other ways and regulation simply pushes them from one type of hedging to another. I provide three pieces of evidence suggesting this is not the case. First, reinsurance is another potential way to manage the risks of VAs (Kojien and Yogo 2022).³ I, however, find no reduction in reinsurance after the regulation changes, which one would expect to see if insurers substitute out of reinsurance and into derivatives. In fact, as I discuss further below, insurers hedge a greater fraction of PRS guarantees with reinsurance after the regulation changes. Second, insurers can hedge the interest rate risk of VA guarantees by buying bonds. However, my analysis of insurers' bond holdings reveals that insurers do not hedge VA guarantees with bonds. This is consistent with the fact that the interest rate risk embedded in VA guarantees is complex and would require complicated dynamic trading to hedge with bonds. Third, I quantify the interest rate exposures of VA guarantees for the insurance sector in the aggregate incorporating all risk management tools. My estimates suggest that the regulatory changes meaningfully closed the risk mismatch for RS: insurers hedge over 80% of RS exposures, largely through derivatives. However, a large fraction (~60%) of PRS exposures remain unhedged.

Overall, my results show that insurers hedge the specific risks that the regulatory regime starts recognizing. An immediate implication is that because the regulatory regime does not yet recognize interest rate risk for PRS, insurers remain exposed to substantial interest rate risk. This is because PRS constitute around two-thirds of VA liabilities and the hedging gap between RS and PRS is large. This inference is consistent with the large sensitivities of insurers' stock returns to interest rates that we see in the data. An important aspect of the new regime is also that it stops recognizing interest rate risk altogether when rates increase. This can further exacerbate insurers' risk mismatch going forward.

³ Using reinsurance insurers can offload risks outside the operating balance sheets either to off-balance-sheet entities called "captives" (shadow insurers) or to third-party reinsurers.

A potential further implication is that prior to the regulatory changes insurers wanted to hedge, but the regulatory framework disincentivized hedging by not properly recognizing the risks of VAs. Prior to 2009, liabilities were largely insensitive, while derivatives were marked-to-market. Thus, insurers properly hedging their economic exposures would have increased the volatility of their regulatory capital. As the regulation changed, hedging incentives were restored, except for PRS guarantees where the regulation does not yet recognize interest rate risk. The evidence on reinsurance discussed above is consistent with this view. We observe insurers employing more reinsurance for PRS relative to RS, which suggests that they want to hedge PRS even though the regulatory regime does not fully incentivize hedging of PRS risks. More broadly, my findings suggest that deficiencies in the regulatory regime can lead financial institutions to bear more risk than they otherwise would.

This paper contributes to several broad strands of recent research on the hedging decisions of financial institutions. First, whether derivatives cause a build up of systemic risk or are used for risk management has been a central question since the financial crisis. Begenu, Piazzesi, and Schneider (2015) show that interest rate derivatives amplify balance sheet fluctuations for U.S. banks. Hoffmann et al. (2019) find the opposite for Euro area banks. This is the first paper to comprehensively measure the risk exposures of the derivatives portfolio of a large subset of the U.S. financial sector over a long period of time and across risk factors. Exploiting granular position-level data, which allows for accurate measurement of risk exposures, I show that insurers use derivatives to hedge the interest rate and equity risks embedded in VA guarantees.

Second, several papers document large heterogeneity in interest rate hedging across banks. The prominent explanations for this heterogeneity include product composition (Hoffmann et al. 2019), collateral constraints (Rampini, Viswanathan, and Vuillemeys 2020), financial frictions (Vuillemeys 2019), costs of distress (Purnanandam 2007), and low interest rate mismatch due to banks' market power in deposit markets (Drechsler, Savov, and Schnabl Forthcoming). To my knowledge, no other paper has examined how regulatory incentives impact risk management decisions and the use of derivatives. I find that differences in the regulatory treatment of products can lead to substantial heterogeneity in the hedging behavior across institutions. Moreover, the detailed data on derivatives allow me to shed light on several novel facets of hedging decisions, including the fraction of exposure hedged, how insurers use different instruments, and how hedging responds to changing macro-conditions.

My paper also contributes to the broader risk management literature on the motives of hedging.⁴ In particular, Smith and Stulz (1985) and DeMarzo and

⁴ The main explanations include (a) market frictions (Smith, Smithson, and Wilford 1990; Stulz 1990; Froot, Scharfstein, and Stein 1993; Smith and Stulz 1985; Froot and Stein 1998) and (b) agency costs (Stulz 1984; Breeden and Viswanathan 1990; Stulz 1990; DeMarzo and Duffie 1992).

Duffie (1992) propose that firms may care about the volatility in accounting exposures, instead of economic exposures if managers' compensation contracts depend on accounting earnings. This may cause hedging to deviate from their economic benchmark. Consistent with these papers, I find that insurers target smoothing the volatility of regulatory capital as a risk management objective.

This paper also closely relates to the broader insurance literature on financial and regulatory frictions and their implications.⁵ In particular, Kojien and Yogo (2022) examine the impact of financial constraints on the VA product market. Kojien and Yogo (2016) study the use of reinsurance as a tool to alleviate regulatory constraints. Ellul et al. (2020) document the impact of miscalibrated risk weights and hedging on asset allocation of VA underwriters. The results on insurers' hedging decisions have consequences for product design, pricing, and market share of VA underwriters. My findings also shed light on why insurers were potentially less well hedged going into the crisis and why reinsurance activities increased, both of which can be traced back to the hedging incentives provided by the regulatory framework.

1. Institutional Setting

This section summarizes the main features of the institutional setting, namely, (a) VA guarantees have significant interest rate and equity exposures and are primarily hedged using derivatives. (b) There are four types of guarantees, and they carry similar economic risks. (c) The regulatory treatment of the guarantees shift in a different manner over time. This gives rise to different hedging incentives across guarantees and risk factors.

1.1 Variable annuities and hedging instruments

Variable annuities (VAs) are long-dated savings products sold by life insurance companies. VAs bundle together mutual funds with long-dated minimum return guarantees, that is, insurers guarantee contract holders a certain minimum rate of return, regardless of the actual performance of the underlying mutual fund. Thus, VAs can be thought of as insurers selling a long-dated put option (guarantees) on the mutual fund with complex payoffs:

$$VA = \text{Mutual Fund} + \underbrace{\text{Long-dated Put Option}}_{\text{Guarantee}}$$

The guarantees expose life insurers to shifts in interest rates and equity markets. When equity markets or interest rates decline, the value of the embedded put option in VA goes up, leading to an increase in insurers'

⁵ See Ellul, Jotikasthira, and Lundblad (2011), Becker and Ivashina (2015), Ellul et al. (2015), and Ge and Weisbach (2021) for asset side implications. Kojien and Yogo (2015a), Ge (2022), and Sen and Humphry (2018) study the effects of regulatory frictions on the product market.

liabilities. VA guarantees, therefore, pose significant risks for insurers, especially in adverse market conditions (Kojien and Yogo 2022).

Hedging instruments. Derivatives are the primary tool that insurers use to manage the risks associated with VA guarantees (Milliman 2021). While the guarantees could also potentially be hedged using securities, practitioners argue that securities are not used (see Section 5). Another prominent tool to manage the risks of VAs is reinsurance (Kojien and Yogo 2022). The empirical tests, therefore, evaluate all potential hedging instruments, namely derivatives, reinsurance, and securities (bonds).

1.2 Types of VA guarantees and similarities in economic exposures

There are four types of VA guarantees. The guarantees are typically referred to as “GMxB,” that is, Guaranteed Minimum Benefit of type “x,” where “x” denotes (a) “A”(accumulation), (b) “D”(death), (c) “I”(income), and (d) “W”(withdrawal). I provide formal descriptions and numerical examples to illustrate how each guarantee works in Appendix A.

Similar economic risk exposures. The four types of guarantees carry similar economic risks. To quantify the risk exposures, I compute the value and the risk sensitivities of the different guarantees numerically as the risk exposures are not directly reported. I use a standard asset pricing model, where the short rate dynamics are given by the Cox-Ingersoll-Ross (1985) model and the equity market dynamics are as under the Black-Scholes model (see Appendix A). For a realistic range of parameter specifications and under various interest rate and equity market conditions, which determine the extent to which the guarantees are in-the-money, I find that all the guarantee types have similar interest rate and equity risk sensitivities (Appendix Table A.2). Thus, shifts in interest rates and the equity market affect the economic value, and therefore the market-value based hedging requirements, of the different guarantee types in very similar ways.

1.3 Regulatory valuation framework and implications for risk management

Next, I describe the two features of the regulatory framework, (1) the treatment of liabilities and (2) the treatment of derivatives, that together give rise to different *regulatory* risk management incentives across guarantees and over time.

1.3.1 Regulatory valuation of liabilities. *Old regime.* Traditionally, the valuation of regulatory liabilities occurred as though the guarantees carried no interest rate risk and limited equity market risk. The regime can be viewed as close to historical cost accounting. Liabilities were insensitive to shifts in interest rates and were discounted using the “valuation rate,” the prevailing interest rate at the time a contract is originated, and not current interest rates. Liabilities also, for the most part, did not reflect equity market movements,

except to some extent large drops in equity market levels.⁶ Thus, to a large extent, the value of regulatory liabilities did not change, from one valuation date to the next, despite changes in interest rates and equity markets.

New regime. A new regulatory framework known as Actuarial Guideline 43 was adopted in September 2008 and implemented in December 2009. The new regime recognized the guarantees' exposures to both interest rate and equity risks by making the valuation of liabilities more risk sensitive relative to the old regime. However, this increase in sensitivities was not applied consistently across all guarantees and risk factors, as I describe below.

Under the new regime, regulatory liabilities are given by the maximum of liabilities computed using two alternative frameworks: *Standard* and *Stochastic*. Under the Stochastic framework, liabilities are computed by simulating the value of the VA guarantees for a range of interest rate and equity market paths, which depend on current market conditions, and then taking an average for the worst 30% of the paths. In contrast, the Standard framework requires insurers to only stress the guarantees for a large instantaneous drop in equity markets *but not changes* in interest rates. Thus, the Stochastic framework is sensitive to both interest rate and equity risks, and the Standard framework is partially sensitive (only to equity risk):

$$\text{Regulatory Liabilities} = \max \underbrace{\text{Standard}}_{\text{Partially sensitive}}, \underbrace{\text{Stochastic}}_{\text{Sensitive}}$$

Crucially, the relative importance of the two frameworks varies for the different guarantees. The Standard framework dominates for some of the guarantees (GMDB and GMIB) with a higher likelihood, while the Stochastic framework dominates for the other guarantees (GMAB and GMWB). Which framework ultimately binds depends on idiosyncratic contract features unrelated to interest rates, as I discuss in Appendix A. For example, the Standard framework is more stringent on mortality risk, which results it in dominating for one type of guarantee but not the others (Junus and Motiwalla 2009). Thus, after the new regulation took effect, the sensitivities of regulatory liabilities diverged for the different guarantees despite economic similarities. Some guarantees became sensitive to *interest rates*, while others remained insensitive. In contrast, *equity risk* sensitivities increased for *all* guarantees.⁷

To simplify the exposition, I henceforth label the VA guarantees where the Stochastic framework dominates as “Risk Sensitive” (**RS**), that is, sensitive to

⁶ In some cases, the old regime required insurers to stress VA guarantees to a drop in equity markets, leading to a rise in liabilities especially when equity markets performed poorly. However, the equity sensitivity under the old regime was smaller than that under the new regime because (a) the equity drop applied only to a portion of the payoffs, and (b) the size of the equity drop was smaller and did not interact with interest rate changes, as is the case under the new regime.

⁷ There are several instances of the regulatory regime of banks and insurers treating products of similar risks differently and products of different risks similarly. See, for example, Coval, Jurek, and Stafford (2009), Acharya and Richardson (2009), Becker and Ivashina (2015), and Becker, Opp, and Saidi (2021).

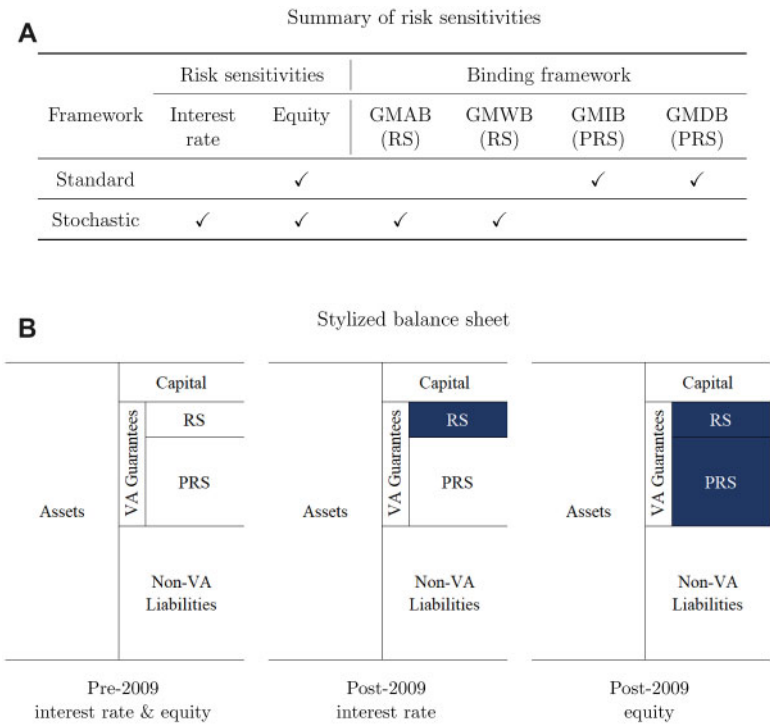


Figure 1
Regulatory valuation framework
Panel A summarizes the risk sensitivities of the regulatory framework and identifies the binding framework for each VA guarantee. Panel B shows stylized balance sheets before and after the shift in the regulation for VA in 2009. Dark shading represents the shift (increases) in the risk sensitivity of the VA guarantees due to the change in the regulation.

both interest rate and equity risk, and where the Standard framework dominates as “Partially Risk Sensitive” (PRS), that is, sensitive to equity, but not to interest rate risk. Figure 1 summarizes the regulatory changes of VA guarantees over time. For completeness, it also depicts the treatment of non-VA liabilities, which continue to be valued under the old regulatory regime.

1.3.2 Regulatory valuation of derivatives. The regulatory regime has always recognized the risks of derivatives, essentially marking them to market. Thus, unlike liabilities, the value of derivatives fluctuates with movements in the underlying risk factors. The disconnect in the treatment of liabilities and derivatives arises because a vast majority of the derivatives (93% by gross notional values) do not qualify for the hedge accounting treatment, which, to a large extent, would have helped to resolve the different treatment that derivatives and liabilities receive under the regulatory regime. To qualify, derivatives must closely replicate the underlying liabilities. This turns out to be

challenging as VAs are long-dated options and the market for traded long-dated derivatives tend to be thin and illiquid, as I discuss in detail in Appendix A. As a result, insurers using derivatives to properly hedge their economic risks would reduce the volatility of their economic capital but generally increase the volatility of their regulatory capital.⁸ Thus, some insurers may have incentives to forgo hedging to the extent that they dislike the volatility of their regulatory capital.⁹

1.3.3 Implications for risk management. If insurers want to reduce the volatility of their regulatory capital, these features of the regulatory framework should give rise to different hedging outcomes across guarantee types, risk factors, and over time. In particular, *interest rate* hedging should increase for RS, but not PRS, guarantees after the regulatory changes, and *equity* hedging should increase for all guarantees. Instead if insurers do not care about the volatility of their regulatory capital, then hedging outcomes across guarantees would be similar as the guarantees have similar economic exposures.

Why might insurers dislike the volatility in regulatory capital and adjust hedging behavior? First, insurance capital regulation requires that insurers maintain a high risk-based capital (RBC) ratio, which is the ratio of the regulatory capital (assets minus liabilities) to required capital. If regulatory capital is too low, insurers may be forced to raise costly external financing. Thus, the presence of RBC constraints make insurers implicitly averse to the volatility in regulatory capital. Second, the volatility in regulatory capital causes fluctuations in the RBC ratio, which could result in more frequent regulatory exams and actions. Third, insurers' credit ratings, which are an important determinant of future demand and product prices, are significantly influenced by the RBC ratio (Kojien and Yogo 2015a). To sustain a certain credit rating, insurers are required to maintain the RBC ratio above a threshold level. Volatility in the RBC ratio may make it more likely that credit ratings are downgraded.

2. Model

There are two dates, $t=0, 1$. The insurer has initial assets A_0 and liabilities L_0 , where assets and liabilities are defined in terms of their respective market values. At time 0, the insurer chooses a hedge portfolio to manage interest rate risk by investing in derivatives. To describe insurers' risk management problem, I solely focus on a single-factor model given by interest rate y .¹⁰

⁸ As the old regime recognized the risks of liabilities in a few states of the world, although to a limited degree, derivatives could potentially bring down the volatility of capital in some cases (see Section 6.2).

⁹ See Credit Suisse (2012). Also, Kojien and Yogo (2022), in discussing why insurers may not fully hedge, suggest that insurers may target the volatility of regulatory capital.

¹⁰ In reality, insurers manage risk exposures to various factors, such as interest rates, credit, and equity risk. The same logic that I develop for interest rates extends to other factors as well.

Interest rate exposures are represented by portfolio durations. At time 0, assets and liabilities have dollar durations β_A and β_L , respectively. By assumption, assets are short-dated relative to liabilities $\beta_A < \beta_L$, thus the insurer has a duration mismatch given by $\beta_G = \beta_L - \beta_A$. The mismatch depends on the level of interest rates and increases when rates fall, that is, $\beta'_G(y) < 0$. This is because VAs embed minimum return guarantees. As interest rates decline, the guarantees become more in-the-money and mismatch increases.

The insurance company wants to hedge the interest rate mismatch, for example, because of financing frictions (Froot, Scharfstein, and Stein 1993), by investing in zero-cost interest rate derivatives with a dollar duration δ_0 , such as swaps.¹¹

The assets and liabilities in market values evolve as:

$$A_1 = A_0 - \beta_A \Delta y - \delta_0 \Delta y, \quad (1)$$

$$L_1 = L_0 - \beta_L \Delta y. \quad (2)$$

The insurer operates subject to “statutory” regulation. The regulatory value of the assets and liabilities evolve as:

$$\hat{A}_1 = \hat{A}_0 - \psi \beta_A \Delta y - \delta_0 \Delta y, \quad (3)$$

$$\hat{L}_1 = \hat{L}_0 - \psi \beta_L \Delta y, \quad (4)$$

where “ $\hat{\cdot}$ ” indicates that assets and liabilities are evaluated as per the regulatory framework. The sensitivities of regulatory quantities are given by $\psi \beta$, where $\psi \in [0, 1]$. $\psi = 1$ implies that the regulatory framework uses market values, while $\psi = 0$ implies that the regulatory framework is insensitive to changes in interest rates. Thus, $\psi < 1$ denotes the wedge between the sensitivity of market values and regulatory values due to the insensitivity of the regulatory framework to shifts in interest rates. Under the regulatory framework, derivatives are marked-to-market, thus the regulatory value of derivatives are not affected by ψ .

Following Koijsen and Yogo (2015a), I define the insurer’s regulatory capital, $\hat{K}_1$, as

$$\hat{K}_1 = \hat{A}_1 - \hat{L}_1 - \phi \hat{L}_1, \quad (5)$$

where $\hat{A}_1 - \hat{L}_1$ denotes the available capital and $\phi \hat{L}_1$ captures the riskiness of liabilities, which can be interpreted as the minimum regulatory capital requirement. $\phi > 0$ implies that regulation is conservative. $\hat{K}$ therefore corresponds to an insurer’s risk based capital.

At time 1, the return on capital, ρ , in market values is given by

$$\rho = \frac{K_1}{K_0} = 1 + (\hat{\beta}_L - \hat{\beta}_A - \hat{\delta}_0) \Delta y \theta_0, \quad (6)$$

where $K_t = A_t - L_t$. I define $\hat{\beta}_L = \beta_L / \hat{K}_0$, $\hat{\beta}_A = \beta_A / \hat{K}_0$, $\hat{\delta}_0 = \delta_0 / \hat{K}_0$, to facilitate comparisons with the empirical tests where I scale dollar durations by

¹¹ In addition to swaps, insurers also use rate options, but in relatively lower quantities.

regulatory capital. Thus, $\theta_0 = \frac{\widehat{K}_0}{K_0}$ emerges because of the change in scaling from economic to regulatory capital.

Equation (6) says that the return on capital at time 1 equals the total interest rate mismatch multiplied by the change in interest rate (Δy).¹²

I define changes in regulatory capital as

$$\frac{\widehat{K}_1}{\widehat{K}_0} = 1 + \left((1 + \phi)\psi\widehat{\beta}_L - \psi\widehat{\beta}_A - \widehat{\delta}_0 \right) \Delta y. \quad (7)$$

The insurer has mean-variance preferences over the return on equity in market values, but faces the costs of regulatory frictions. I model the costs of regulatory frictions through a convex cost function, $C(\widehat{K})$, which captures the costs of low regulatory capital, as in Kojien and Yogo (2015a). Thus, the insurer chooses its hedge portfolio in derivatives according to

$$\max_{\widehat{\delta}_0} \quad \mathbb{E}(\rho) - \frac{\xi}{2} \text{Var}(\rho) - \mathbb{E}[C(\widehat{K})], \quad (8)$$

where ξ captures the insurer's aversion to volatility in the return on equity. Equation (8) highlights two separate sources of volatility aversion. The second term captures the aversion to volatility in economic capital. The first two terms can be interpreted as a reduced form to capture the trade-offs in Froot, Scharfstein, and Stein (1993), where firms maximize the expected profits subject to costly external finance, and are therefore implicitly averse to volatility in profits. The third term arises because volatility in regulatory capital can induce additional costs over and above the volatility in economic capital, for example, because of regulators forcing insurers to raise equity at times they would not do otherwise,¹³ frequent regulatory exams, or a rating downgrade. For tractability and ease of interpretation, I define the expected regulatory cost to be proportional to its variance:

$$\mathbb{E}[C(\widehat{K})] = \frac{\gamma}{2} \text{Var}\left(\frac{\widehat{K}_1}{\widehat{K}_0}\right) \lambda(\widehat{K}_0), \quad (9)$$

where $\gamma\lambda(\widehat{K}_0) \geq 0$ parameterizes the importance of regulatory constraints. I assume $\lambda'(\widehat{K}_0) < 0$, implying that weakly capitalized insurers are more averse to fluctuations in regulatory capital.¹⁴

¹² I ignore the dependence of duration on y as the insurer only duration hedges by assumption. The insurer, therefore, ignores the volatility in capital that arises due to shifts in duration at $t=1$ due to shifts in y .

¹³ For example, suppose an insurer perfectly immunizes its economic capital with derivatives. If rates were to rise, the value of the hedge portfolio could decline substantially without an offsetting decline in regulatory liabilities as they are rate insensitive. As a result, regulatory (and not economic) capital would decline forcing insurers to raise equity in states of the world they would not do otherwise.

¹⁴ An alternative approach is to explicitly add a regulatory constraint. This, however, yields a similar reduced form. For example, Equation (9) is observationally equivalent to a Value-at-Risk constraint, written as proportional to the variance of changes in regulatory capital (Adrian and Shin 2014; Miranda-Agrippino and Rey 2020), where $\lambda(\widehat{K}_0)$ is the Lagrange multiplier on the constraint. Intuitively, as an insurer would care about all future constraints that could bind, volatility of regulatory capital would matter for hedging.

The first-order condition yields the following solution for optimal hedging:

$$\widehat{\delta}_0 = \frac{-\mathbb{E}(\Delta y)\theta_0}{(\gamma\lambda(\widehat{K}_0) + \xi\theta_0^2)\sigma^2} + (\widehat{\beta}_L - \widehat{\beta}_A) \frac{\gamma\psi\lambda(\widehat{K}_0) + \xi\theta_0^2}{\gamma\lambda(\widehat{K}_0) + \xi\theta_0^2} + \phi\psi\widehat{\beta}_L \frac{\gamma\lambda(\widehat{K}_0)}{\gamma\lambda(\widehat{K}_0) + \xi\theta_0^2}, \quad (10)$$

where σ^2 is the volatility of interest rate shocks. I prove Equation (10) in Appendix B. Equation (10) implies that the optimal demand for derivatives consists of three terms. The first term is due to a speculation motive, which depends on the expected changes in interest rates, $\mathbb{E}(\Delta y)$, the volatility of interest rate shocks, and total risk aversion, $\gamma\lambda(\widehat{K}_0) + \xi\theta_0^2$. The second and third term are due to a hedging motive and depend on the duration gap, $\widehat{\beta}_L - \widehat{\beta}_A$, the wedge between market and regulatory values, ψ , and capital requirements, ϕ .

Implication 1: When the expected regulatory cost is low, and assuming the insurer does not use derivatives for speculation, $\mathbb{E}(\Delta y)=0$, there is full *economic* hedging:

$$\lim_{\gamma\lambda(\widehat{K}_0) \rightarrow 0} \widehat{\delta}_0 = (\widehat{\beta}_L - \widehat{\beta}_A). \quad (11)$$

When the expected regulatory cost is high, there is full *regulatory* hedging:

$$\lim_{\gamma\lambda(\widehat{K}_0) \rightarrow \infty} \widehat{\delta}_0 = \psi(\widehat{\beta}_L - \widehat{\beta}_A) + \phi\psi\widehat{\beta}_L. \quad (12)$$

Equation (11) yields the baseline hedging without regulatory costs and a speculation motive. In the baseline, the insurer minimizes the volatility in economic capital by fully hedging the interest rate mismatch between assets and liabilities.¹⁵ Equation (12), on the other hand, implies that when regulatory costs are sufficiently high, the insurer only cares about the volatility in regulatory capital, which depends on ψ and ϕ , in addition to interest rate mismatch, resulting in a wedge between economic and regulatory hedging.

Implication 2: *The effect of a risk insensitive regulatory framework:* The optimal hedging demand increases with the risk sensitivity of the regulatory framework ψ , as $\frac{\partial \widehat{\delta}_0}{\partial \psi} > 0$.

Intuitively, insurers minimize the volatility of changes in regulatory capital in order to reduce the expected regulatory cost. The volatility is higher when the insurer hedges than when it does not, as only derivatives are sensitive to interest rates, while the rest of the regulatory balance sheet is insensitive, or only partially sensitive. Thus, when ψ is low, hedging is lower.

Implication 3: *The effect of conservative capital regulation:* The optimal hedging demand increases due to conservative regulation $\phi > 0$, as $\frac{\partial \widehat{\delta}_0}{\partial \phi} > 0$.

Thus, as regulation becomes more conservative and regulatory capital requirements increase, that is ϕ goes up, hedging increases as well.

¹⁵ The choice of full hedging as a baseline is for simplicity. The literature identifies various reasons for incomplete hedging (e.g., net worth constraints), which the model leaves out. Section 6 shows relatively high hedging in the absence of regulatory frictions, suggesting that insurers want to hedge a large fraction of risk exposures and using full hedging as a baseline is a reasonable starting point.

2.1 Testable predictions

The shift in the regulatory framework resulted in an increase in ψ as some VA guarantees became more sensitive to interest rate changes. Moreover, due to the differences in the amount of RS guarantees underwritten, the shift in the regulation resulted in heterogeneity in ψ across insurers. That is, insurers that have more RS guarantees have a higher ψ . This gives the following predictions, which I test using detailed contract level data on derivatives.

Prediction 1: *Magnitude of hedging*: Over time, an increase in ψ leads to an increase in hedging, all else being equal, as shown in implication 2. Moreover, for two firms, i and j , all else being equal, if $\psi_i > \psi_j$ then $\hat{\delta}_{0i} > \hat{\delta}_{0j}$. Thus, the impact of the shift in the regulation on hedging demand is higher for firms with a higher ψ .

Prediction 2: *Dynamics between hedging and interest rates*: An increase in ψ implies a higher sensitivity of hedging demand to interest rates, all else being equal, as $\frac{\partial^2 \hat{\delta}_0}{\partial \psi \partial y_0} < 0$. In other words, if interest rates fall, hedging demand increases, but the increase is higher when ψ is higher. Moreover, for two firms, i and j , all else being equal, if $\psi_i > \psi_j$ then $\frac{\partial \hat{\delta}_{0i}}{\partial y_0} < \frac{\partial \hat{\delta}_{0j}}{\partial y_0}$. Thus, firms with a higher ψ , adjust hedging demand more in response to changes in interest rates, than firm with lower ψ , all else being equal.

Prediction 3: *Interaction with regulatory constraints*: (A) As regulatory constraints become more binding ($\lambda(\hat{K}_0)$ increases), the greater is the marginal effect of a shift in ψ as $\frac{\partial^2 \hat{\delta}_0}{\partial \lambda(\hat{K}_0) \partial \psi} > 0$. Intuitively, when the regulatory constraints are more binding, the volatility of regulatory capital is more costly. If the regulatory framework is risk insensitive, this would lead to lower hedging because using more derivatives would have increased the volatility of regulatory capital. As the risk sensitivity of the regulatory framework increases, it would generate a larger increase in hedging for insurers that were ex ante more capital constrained as the risk insensitivity of the framework affected these insurers more. (B) Moreover, regulatory constraints affect hedging in opposite ways depending on the magnitude of ψ . When $\psi = 0$, $\frac{\partial \hat{\delta}_0}{\partial \lambda(\hat{K}_0)} < 0$. Thus, an increase in the importance of regulatory constraints leads to a decrease in the hedging demand. However, when $\psi = 1$, $\frac{\partial \hat{\delta}_0}{\partial \lambda(\hat{K}_0)} > 0$. Thus, an increase in the importance of regulatory constraints leads to an increase in hedging demand due to conservative regulation ϕ . I prove prediction 3 in Appendix B.

3. Data and Measurement

I combine position-level data on interest rate and equity derivatives with data on VA guarantees, regulatory capital ratios, and other insurers' characteristics to examine the impact of differences in regulatory treatment on hedging outcomes.

3.1 Derivatives holdings data

Life insurers report quarterly positions and transactions in derivatives, which I collect from the NAIC's Schedule DB database. The data on individual derivatives positions are unique to the literature on risk management for several reasons. First, every single position for each life insurer is reported, including all the instruments held, across all the risk factors hedged, and over a long period of time, starting in 2004, which allows me to study the evolution of hedging over various interest rate, equity market, and regulatory regimes.¹⁶ Second, detailed contract characteristics are available for each position, which allows me to compute the risk exposures of derivatives in a consistent way for each insurer and across multiple risk factors. Third, the data contain other important aspects of hedging at the position level, including a description of which balance sheet item is being hedged, how effective are the hedges (hedge accounting), and counterparties and collaterals posted. In aggregate, over the entire sample from 2004 to 2016, there are 2.6 million derivative positions, equaling over \$60 trillion in notional value. The relevant data features are described in more detail below.

3.1.1 Risk factors. The major risk factors hedged are interest rate, equity, currency, and credit risk. For each position, starting in the year 2010, insurers report which risk factor is being hedged. However, before 2010, risk factors are not directly reported but can be discerned from the position descriptions, which I extract using textual analysis (see Appendix E). Figure 2 shows a breakdown of the gross notional values by risk factors. Interest rate derivatives are by far the largest, followed by equity derivatives. Moreover, interest rate and equity derivative holdings have been significantly increasing since 2009, when the regulatory regime changed, unlike the holdings of the other risk factors. I, henceforth, focus on interest rate and equity risks as these are the main financial risks that VA are exposed to.

3.1.2 Instruments and contract characteristics. The main derivative instruments include swaps, options, futures, and forwards. For interest rate risk, I restrict my analysis to swaps and options only, as these comprise more than 95% of all contracts by gross notional values.¹⁷ For equity risk, I restrict attention to futures and put options as the bulk of these instruments are earmarked in the data as hedging VA guarantees. I only consider positions

¹⁶ Alternative data sources (e.g., annual reports) have lower coverage of firms as many insurers are "mutuals" (i.e., a private company, see Appendix F) and often include the holdings of other subsidiaries (e.g., Property & Casualty or Asset Management).

¹⁷ Kirti (2017) examines the risk-taking behavior of U.S. insurance companies and uses positions data on interest rate derivatives, focusing on interest rate swaps.

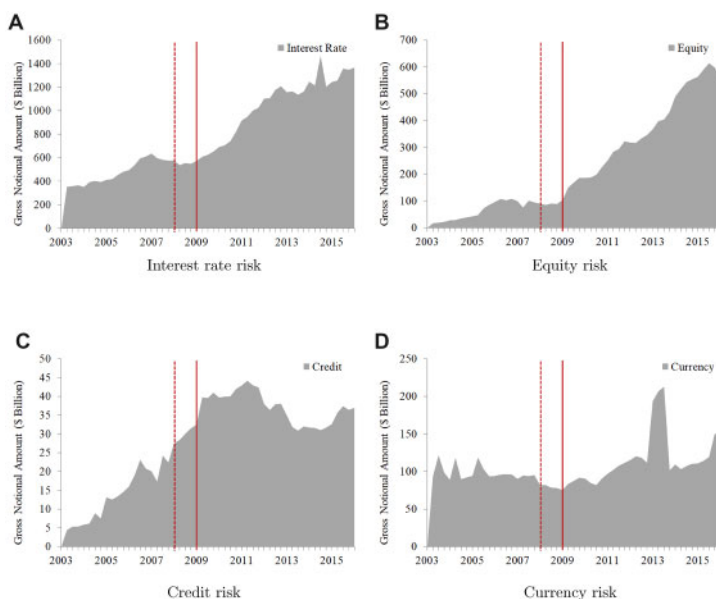


Figure 2
Risks hedged by derivatives

This figure shows a breakdown of the total gross notional values by risk factors hedged for life insurers in the aggregate. I plot gross notionals, aggregated across instruments, for each risk factor from 2004 to 2016 at a quarterly frequency. Interest rate risk is the largest hedged risk factor, followed by equity risk. Together, the two risk factors account for over 90% of the total positions by gross notionals in Q4:2016. The solid vertical line (dotted line) represents the implementation (adoption) of the new regulation. The start date is dictated by the availability of the derivatives data.

where the underlying is the S&P 500, which represent over 70% of all equity derivatives by gross notional amount.¹⁸

The main advantage of the data is that detailed contract characteristics are available for each position, including direction, maturity, strike, underlying, fair values, and trading data. Thus, as opposed to using the gross notional amounts, which may not reflect the true extent of hedging and the actual risk exposures, I can measure hedging more accurately by aggregating up the exposures of individual positions.¹⁹ Some contract characteristics get reported in position description strings. I use textual analysis or the reported fair values to infer the relevant contract features in such cases (see Appendix F). Overall, contract characteristics are available for 92% of the positions by gross notional values (see Appendix Table F.2).

¹⁸ For equity risk, the major instruments include calls, puts, and futures. Call options are mainly earmarked for replicating equity indexed contracts. As the analysis focuses on VA, I exclude the data on call options.

¹⁹ Gross notionals may not accurately measure hedging because positions may have (a) opposite directions and risk exposures (e.g., receive or pay fixed swap), which do not get netted out, and (b) different maturities and moneyness, but all positions receive equal weight regardless of actual exposures.

Table 1
Descriptive statistics: A snapshot of derivative positions

<i>A. Interest rate derivatives</i>									
Maturity (years)	Total	Rate decrease				Rate increase			
		≤ 1	2-5	6-9	≥ 10	≤ 1	2-5	6-9	≥ 10
Swaps	868	26	116	66	274	27	85	80	192
Caps	235	0	2	2	0	50	141	40	0
Floor	17	5	1	7	0	4	0	0	0
Swaptions	47	16	10	1	3	8	6	1	3

<i>B. Equity derivatives</i>									
Maturity (years)	Total	Equity fall				Equity rise			
		≤ 1	2-5	6-9	≥ 10	≤ 1	2-5	6-9	≥ 10
Puts	86	53	16	5	1	8	2	1	0
Futures	32	25	0	0	0	7	0	0	0

The table shows a snapshot of derivative positions for the insurance sector as a whole as of Q4 2016. I break the positions into (a) *risks hedged*: interest rate derivatives (panel A) and equity derivatives (panel B); (b) *instruments*: swaps, caps, floors, and swaptions for interest rate derivatives and puts and futures for equity derivatives; (c) *trade direction*: rate decrease (increase) and equity fall (rise) shows positions that gain in value when interest rates decrease (increase) and the equity market falls (rises); and (d) *maturity* (in years), which is split as less than 1 year, 2 to 5 years, 6 to 9 years, and greater than 10 years. I report gross notional for interest rate derivatives (in billions of dollars) and the total number of contracts for equity derivatives (in millions).

Table 1 shows a snapshot of the data as of 2016:Q4. There is significant heterogeneity in the types of instruments for both interest rate and equity derivatives. For example, within the interest rate positions, 65% are swaps and 35% are options (caps, floors, and swaptions). Furthermore, there is also significant heterogeneity in contract features, including maturity and the direction of exposure. I convert each position into its corresponding risk exposures, fully accounting for the heterogeneity in the contract features (see Section 4).

3.2 Which liabilities are derivatives hedging?

Next, I turn to the cross-sectional distribution of derivative holdings, which show that derivatives are predominantly concentrated in insurers that sell VA. First, to assess to what extent insurers use derivatives to hedge the different types of liabilities, I collect data on insurers' liabilities from statutory filings. The main insurance liabilities are Life and Annuities. Annuities are further broken into Variable Annuities Guarantees (G), Fixed Annuities (FA), and Indexed Annuities (IA). I regress the gross notional amounts on the various liability types, controlling for other insurer characteristics (e.g., total assets and leverage). Appendix Table C.1 shows a positive relationship between the size of derivative holdings and the amount of VA guarantees; that is, insurers that have more VA also hold more derivatives.²⁰ For example, as we move from the

²⁰ The sample starts in 2010, when a breakdown of annuities into FA and IA becomes available. However, the results look extremely similar in the full sample using the combined non-VA annuity as the regressor.

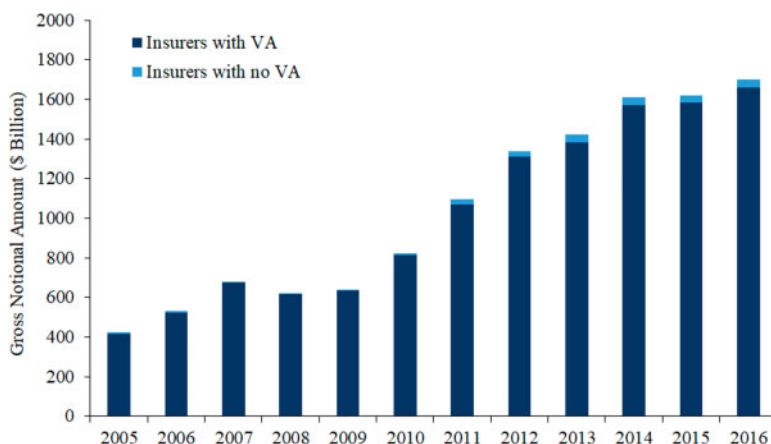


Figure 3
Who holds derivatives?

This figure shows a breakdown of the total gross notional amounts by whether or not insurers have VA guarantees as liabilities during a particular year. I plot gross notional amounts, aggregated across instruments, for interest rate and equity derivatives. Insurers *with VA* are insurers that had at least some VA guarantees as liabilities and insurers *with no VA* had no VA guarantees as liabilities in a particular year. The sample is 2005 to 2016. The start date and frequency are dictated by the availability of the data on VA guarantees.

25th to the 75th percentiles of the distribution of VA guarantees, gross notionals in interest rate derivatives go up more than 10 times from 7 cents to 73 cents per dollar of capital. Similarly, gross notionals in equity derivatives go up from 3 cents to 32 cents per dollar of capital. In contrast, there is negligible relationship between the size of derivative holdings and other types of liabilities, including life, FA, and IA.²¹

Second, consistent with the cross-sectional evidence, the aggregates also show that derivatives are primarily held by insurers that sell VA guarantees. Figure 3 shows a breakdown of the gross notional amounts by whether or not insurers have VA guarantees as liabilities during a particular year. Insurers that have VA hold 98% of all derivatives on average across years. In other words, derivatives are concentrated in roughly the 25% of insurers that sell VA guarantees at some point. Overall, these facts imply that derivatives are primarily hedging the VA portion of insurers' liabilities. Could insurers also use bonds to hedge the interest rate risk in VA? In Section 5, I show that is not the case and, therefore, the main analysis focuses on derivatives. Next, I describe the VA data in greater detail.

²¹ Results hold both at univariate and multivariate levels, implying that correlation between G and other liabilities is not driving the estimates. Moreover, consistent with the risk management literature, large insurers hold more derivatives, but as my results show, product composition matters in addition to firm size.

3.3 Variable annuities data and measurement of regulatory risk sensitivities

Insurers report gross liabilities, amounts reinsured, and the account values for each VA guarantee at an annual frequency starting in the year 2005 in their statutory filings.²² Gross liabilities measure the regulatory value of the guarantees underwritten, that is, the value of the embedded put option within a VA contract. Amounts reinsured are the liabilities ceded to another entity, which helps move risk exposures from the balance sheet of the insurer to a different entity. Account values measure the value of the mutual fund part of the VA contract. For each guarantee, I compute *net liabilities*, that is, the gross liabilities minus the amounts reinsured. Thus, the net liabilities measure the portion of the liabilities that still remain on the insurer's balance sheet and affect the sensitivity of regulatory liabilities and capital. Table 2 shows descriptive statistics of gross liabilities, reinsurance, and net liabilities for the different types of VA guarantees as of 2016:Q4.

The new regulatory framework resulted in a shift in the regulatory risk sensitivities of VA guarantees that have similar economic risk exposures. I measure this shift by the ex-ante ratio of VA guarantees to total capital and surplus (shortened to capital henceforth). I denote the ratio of total VA guarantees to capital by G_i . I split this into two parts: the ratio of risk sensitive guarantees to total capital, RS_i , and the ratio of partially risk sensitive guarantees to total capital, PRS_i . Specifically, I define:

$$\begin{aligned}
 G_i &= \frac{(\text{Net Liabilities in VA Guarantees})_{i,2007}}{(\text{Regulatory Capital})_{i,2007}} \\
 &= \frac{(\text{Net RS Liabilities} + \text{Net PRS Liabilities})_{i,2007}}{(\text{Regulatory Capital})_{i,2007}} \quad (13) \\
 &= RS_i + PRS_i,
 \end{aligned}$$

where the components of RS and PRS guarantees are defined in Section 1.3. Table 2 shows the distributions of RS_i and PRS_i . A higher RS_i implies that regulatory capital became more sensitive to both interest rate and equity market movements after 2009. In contrast, a higher PRS_i implies that regulatory capital became more sensitive to the equity market, but not to interest rate movements after 2009.

I define these ratios as of year-end 2007 for two reasons. First, I want to avoid the endogeneity concern that companies simultaneously choose their exposure in these products that are somehow correlated with the change in the regulation itself. Second, insurers report the VA liabilities after accounting for

²² See, section 9.2 General Interrogatories Part 2 of the regulatory financial statements.

Table 2
Descriptive statistics: VA guarantees and firm characteristics

A. VA guarantees							
	Mean	SD	P10	P25	Median	P75	P90
Total liabilities & reinsurance (\$ billion)							
Total (gross)	1.28	2.94	0.00	0.01	0.13	0.86	3.94
RS (gross)	0.08	0.17	0.00	0.00	0.01	0.10	0.27
PRS (gross)	1.19	2.93	0.00	0.00	0.09	0.48	3.28
Total (net)	0.56	1.58	0.00	0.00	0.04	0.33	1.02
RS (net)	0.04	0.08	0.00	0.00	0.01	0.04	0.11
PRS (net)	0.52	1.57	0.00	0.00	0.03	0.24	0.73
Reinsured: Total (%)	32	39	0	0	5	72	96
Reinsured: RS (%)	20	34	0	0	1	35	94
Reinsured: PRS (%)	33	40	0	0	6	77	95
Ex ante RS and PRS ratios (%)							
Total ratio: G_i	4.0	8.1	0.0	0.0	0.4	4.1	15.5
RS ratio: RS_i	0.8	2.2	0.0	0.0	0.0	0.3	2.5
PRS ratio: PRS_i	3.2	7.7	0.0	0.0	0.2	1.7	10.2
B. Firm characteristics							
	Mean	SD	P10	P25	Median	P75	P90
Total assets (\$ billion)	120	134	20	29	59	179	291
Capital & surplus (\$ billion)	7	8	1	2	5	9	20
Leverage	10.2	4.2	5.0	7.9	9.4	12.8	15.8
RBC ratio	9.8	2.4	6.9	8.2	9.9	11.3	12.3
Rating	2.7	1.0	1.0	2.0	2.7	3.0	4.0
BCAR - guideline	129	143	29	54	93	180	276
Shock	0.92	0.20	0.64	0.81	0.95	1.02	1.18
Separate account share (%)	31	26	0	10	31	58	70
Generosity (%)	3	14	0	0	0	1	3
MBS share (%)	21	11	7	14	20	27	33
Callable share (%)	32	22	7	18	26	54	68

Panel A shows descriptive statistics on VA guarantees. Total liabilities and reinsurance data are as of Q4:2016. *Gross liabilities* are total liabilities underwritten. *Net liabilities* are liabilities after reinsurance. *Reinsured* is the proportion of gross liabilities that is reinsured. *Ex ante RS and PRS ratios* are the ratio of net RS and PRS liabilities to total capital as of 2007 (see Equation (13)). Panel B provides descriptive statistics as of Q4:2016 on other characteristics. (a) *Total assets* are admitted assets; (b) *Capital and surplus* measures total available capital; (c) *Leverage* is the ratio of total general account assets to total available capital; (d) *RBC ratio* is the ratio of total available capital to total required capital; (e) *Rating* is the AM Best rating converted to a numeric scale from 1 (A++) to 15 (F); (f) *BCAR minus guideline* is the difference between AM Best's capital adequacy ratio and the guideline ratio to sustain the current rating; (g) *Shock* is the change in capital between 2007:Q4 and 2009:Q1; (h) *Separate account share* is the total account value in VA divided by total liabilities; (i) *Generosity* is the ratio of net liabilities in VA guarantees divided by the total account value; and (j) *MBS and callable share* is the total fair value of MBS and callable bonds divided by the total fair value of the bond portfolio. The sample includes insurers that hold interest rate derivatives.

hedging. Thus, ex post it is unclear whether high liabilities are due to high prior exposures or low hedging. As the regulatory framework was adopted in 2008 and because insurers hedged little before 2008-2009 (see Section 4), using liabilities as of 2007 helps to circumvent both issues. The identification strategy exploits the cross-insurer variation in the ex ante ratio of the different guarantee types to total capital (RS_i and PRS_i) to trace hedging across insurers, over time, and for each risk factor (interest rate and equity). The exact empirical specifications are discussed in Section 4.

3.4 Insurers' characteristics and sample

Regulatory constraints. The model shows that when the regulatory constraints are more binding, greater volatility of regulatory capital is more costly. To measure regulatory constraints, I collect data on the risk-based capital (RBC) ratio from insurers' statutory filings, which are reported at an annual frequency. A high RBC ratio indicates that an insurer is well capitalized relative to its capital requirements.

Additional characteristics. The risk management literature provides several explanations for why hedging could vary across firms and over time. The main explanations include financing frictions (Froot, Scharfstein, and Stein 1993); ability to hedge and pay fixed hedging costs (Nance, Smith, and Smithson 1993); and net worth (collateral) constraints (Rampini and Viswanathan 2010). Measures of firm size and financing constraints proxy for these factors. I measure size using $\log(\text{assets})$. Measures of financing constraints include leverage (general account assets divided by total capital), A.M. Best's credit ratings, capital adequacy ratio (BCAR) ratio, and the guideline BCAR that sustains current ratings.

Sample. I aggregate balance sheets and derivative holdings at the insurance group level using NAIC company code to group code mappings. The sample includes insurers that held derivatives during the period 2004–2016. For example, 57 insurers held interest rate derivatives at some point during the sample period, which are predominantly insurers that sell VA guarantees (Figure 3). As I compare hedging before and after the shift in the regulatory framework, I only include insurers that have been in operation for at least 2 years in each regulatory regime. I also exclude insurers that have less than \$100 million gross notional in rate derivatives combined across all quarters. The final sample contains 53 insurers, which account for 89.3% of the sector's total assets at the end of 2016. I discuss the descriptive statistics and the main empirical findings in the next section.

4. Regulatory Incentives and Hedging with Derivatives

4.1 Measurement of risk exposures of derivative holdings

I measure the extent of hedging by computing the risk exposures of the derivative holdings. Specifically, I compute the interest rate exposures for each position due to a 100-bps shift in the yield curve. I measure the risk exposures of swaps by dollar duration and of rate options by option "deltas" as implied by the Black (1976) model. Thus, I assume parallel shifts of the yield curve, which as term structure decompositions show, account for a vast majority of the variation in yields.²³ I compute the equity risk exposures due to a 10%

²³ Litterman and Scheinkman (1991) show that most of the variation in returns of fixed income securities can be explained by three factors: level, slope, and curvature, where the level factor corresponds to a roughly parallel shift of the yield curve. See also Cochrane and Piazzesi (2005).

shift in the S&P 500 index. The exposures of futures are +1 for long and -1 for short positions for a \$1 change in S&P 500, and for put options, I compute the Black-Scholes deltas.²⁴

The measurement of risk exposures is a three-step process. First, I compute the exposures of a wide range of instruments that have different direction, maturity, and moneyness at the end of each quarter t given current interest rate and equity market conditions (e.g., zero and forward curves, the level of the S&P 500, and implied volatilities).²⁵ Second, I apply the exposures to the position-level data by matching the relevant contract features. For example, for swaps, I match the direction (receive or pay fixed), remaining maturity, and the swap rate, and for options, I match the strike, maturity, and the trade direction. Appendix F describes the methodology and the data steps. Finally, I compute the total dollar risk exposures δ_{it} for insurer i at time t for each risk factor by aggregating exposures across positions as follows:

$$\delta_{it} = \sum_j \Delta_{jt} \times N_{ijt}, \quad (14)$$

where Δ_{jt} is the exposure of position j at time t .²⁶ For each instrument, I compute exposures for a hypothetical \$1 in notional value (interest rates) or a \$1 change in the S&P 500 (equity). I then multiply exposures by N_{ijt} , which is the notional value in the case of interest rate positions or the number of contracts in the case of equity positions.

I aggregate the risk exposures across positions for the following reasons. First, some positions may have opposite directions and risk exposures (e.g., receive fixed vs. pay fixed swaps), aggregation allows the offsetting positions to net out. Moreover, positions that have higher exposures (e.g., long maturity swaps or in-the-money options) receive a greater weight in the aggregation relative to positions that have lower exposures. Thus, δ_{it} measures what would be the approximate total *net* change in the value of the derivative holdings if interest rates were to shift by 100-bps or equity markets by 10%. Figures 4 and 6 show the time series of the total exposures from interest rate and equity derivatives, respectively, for the insurance sector as a whole. Appendix F validates the accuracy of the exposure calculations by matching the joint dynamics of interest rate changes, calculated exposures, and the ex post reported changes in the value of insurers' derivative holdings.

²⁴ Begenau, Piazzesi, and Schneider (2015) measure risk exposures using factor regressions, which account for nonparallel movements in yields as well. My data contain a substantial portion of options, where a factor regression can be challenging to implement due to a lack of reliable returns data.

²⁵ Appendix Figure F.1 and Table F.1 show exposures of standard 10-year instruments as examples.

²⁶ The analytical expressions for computing Δ_{jt} can vary as there are different instruments in the data. Appendix F describes these expressions. Δ_{RFS} and Δ_{PFS} denote the exposures of swaps, which are depicted in Equations (E.3) to (E.6). Δ_C , Δ_{FL} , Δ_{PS} , and Δ_{RS} denote the exposures of caps, floors, payer swaptions, and receiver swaptions, which are depicted in Equations (E.8) to (E.11), respectively.

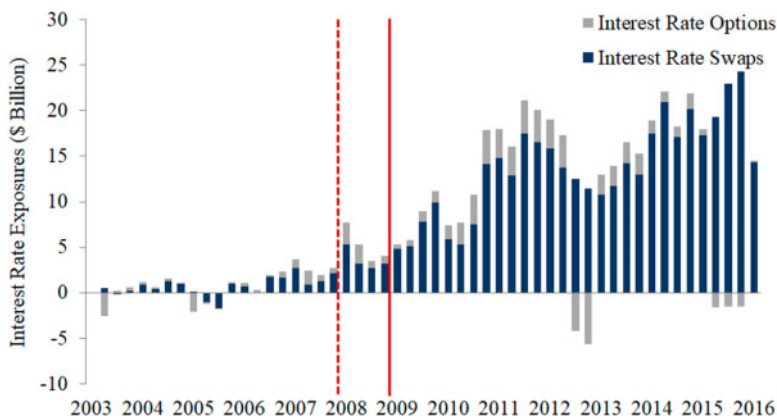


Figure 4
Interest rate risk exposures from derivatives

This figure shows the total dollar risk exposures from interest rate derivatives due to a 100-bps parallel shift in the yield curve. I incorporate all contract characteristics, including direction of the position, maturity, and moneyness in computing the risk exposures (see Section 4.1). The dark bars represent exposures from swaps and the light bars represent exposures from options. Positive (negative) exposures imply that the derivative portfolio gains (loses) in value when interest rates fall. The solid vertical line (dotted line) represents the implementation (adoption) of the new regulation. Exposures are aggregated for all the insurers in the sample that hold interest rate derivatives. The data frequency is quarterly from 2004:Q1 to 2016:Q4, and the start date is dictated by the availability of the derivatives data.

4.2 Interest rate risk hedging

4.2.1 Stylized facts. I start by documenting stylized facts on interest rate risk hedging. First, interest rate derivatives have a *positive* duration on average, which implies that derivatives gain in value when interest rates decline (Table 3). Insurers are exposed to declines in interest rates because VA embed minimum return guarantees. As interest rates decline, these guarantees become more in-the-money and the value of the liabilities increase. A positive duration of derivatives, therefore, implies that insurers buy protection, that is, hedge declines in interest rates to offset the increase in liabilities. Consistent with this, Appendix Figure A.1 shows that 96% of the positions are reported as being held for hedging and not speculation.

Second, Figure 4 shows that interest rate exposures of derivatives have increased substantially since 2009. A 100-bps parallel shift in the yield curve would have resulted in a negligible change in the value of the interest rate derivatives portfolio before 2009, however, at its peak in 2016, the same shift in interest rates would have led to a change of over \$22 billion, or 8% of total capital, for insurers as a whole.²⁷ Table 3 shows that the average insurer that

²⁷ δ_{it} can shift due to shifts in Δ_{jt} , as exposures change with interest rates, and shifts in N_{ijt} , that is, with changes in actual positions. Thus, I construct an alternative measure where I fix Δ_{jt} as of 2010:Q2, which is the date closest to the average 10-year yield during my sample. I find that exposures have increased primarily due to changes in actual positions.

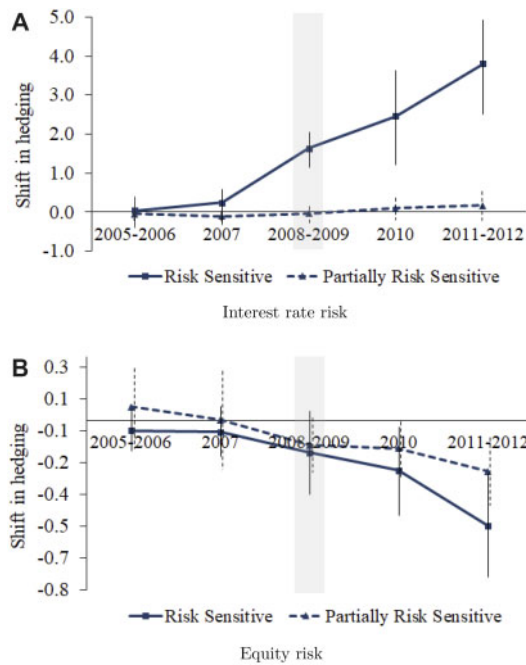


Figure 5
Regulatory incentives and shift in hedging exposures

This figure shows the dynamics of the shift in hedging for the RS and PRS ratios immediately around the regulatory change. I regress the main dependent variables, interest rate hedging exposures scaled by capital (panel A), and equity hedging exposures scaled by capital (panel B) on the interaction term between RS and PRS ratios and time dummies for 2-3 years before, 1 year before, during, 1 year after, and 2-3 years after the regulatory change. The figure plots the coefficients and the 95% confidence intervals. The base year is 2004. The gray bar represents the adoption (2008) and implementation (2009) period. The sample includes insurers that hold interest rate or equity derivatives. Frequency of the data are annual.

holds derivatives has had more than a sixfold increase in risk exposures during the period 2009-2016 relative to the period 2004-2008.

Third, while the extensive margin, that is, whether or not insurers hold derivatives, is correlated with whether or not insurers have VA guarantees (Figure 3; Appendix Table C.1), the intensive margin is primarily associated with the *composition* of the VA guarantees. Conditional on holding derivatives, I estimate a cross-sectional mean specification, in which I regress the time-series average of derivatives exposures and volatility of derivatives exposures for each insurer on the average RS ratio, PRS ratio, total assets, leverage, and RBC ratio:

$$\overline{Y}_i = \alpha + \beta_1 \overline{RS}_i + \beta_2 \overline{PRS}_i + \beta_3 \overline{\text{Log}(\text{assets})}_i + \beta_4 \overline{\text{Leverage}}_i + \beta_5 \overline{RBC}_i + \epsilon_i,$$

where $\overline{Y}_i$ is $\widehat{\delta}_i$, the average derivatives exposures scaled by capital, or $\sigma(\widehat{\delta}_i)$, the standard deviation of derivatives exposures scaled by capital. Table 4 shows that insurers with a high RS ratio also have high interest rate exposures.

Table 3
Descriptive statistics: Cross-sectional distribution of hedging

<i>A. Interest rate risk</i>							
	Mean	SD	P10	P25	Median	P75	P90
2004-2008							
Gross notionals (\$ billion)	8.01	15.03	0.00	0.02	0.35	9.43	22.25
Exposures (\$ billion)	0.04	0.22	-0.08	0.00	0.00	0.01	0.16
Exposure/ Capital	0.5	4.0	-0.8	0.0	0.0	0.4	2.1
2009-2016							
Gross Notionals (\$ billion)	16.73	32.29	0.01	0.15	0.93	18.90	53.64
Exposures (\$ billion)	0.26	0.50	0.00	0.00	0.01	0.36	0.84
Exposure/ Capital	3.4	6.9	-0.3	0.0	0.5	4.3	9.7
<i>B. Equity risk</i>							
	Mean	SD	P10	P25	Median	P75	P90
2004-2008							
No. of contracts (million)	0.37	1.47	0.00	0.00	0.07	0.19	0.37
Exposures (\$ million)	0.78	52.10	-24.48	-6.34	0.00	0.00	13.06
Exposure/ Capital	0.23	1.97	-0.41	-0.09	0.00	0.00	0.43
2009-2016							
No. of contracts (million)	2.33	6.11	0.00	0.00	0.16	1.53	6.56
Exposures (\$ million)	-88.2	221.4	-298.4	-54.6	-13.7	-0.09	2.03
Exposure/ Capital	-1.37	2.95	-5.92	-1.00	-0.36	0.00	0.11

The table provides descriptive statistics on interest rate (panel A) and equity derivative (panel B) positions, including gross notionals, exposures, and exposures per unit of capital for interest rate and the total number of contracts, exposures, and exposures per unit of capital for equity positions. I compute averages for each insurer separately for the two regulatory regimes, 2004-2008 and 2009-2016, using quarterly data and then report the cross-sectional distribution. The sample includes insurers that hold interest rate or equity derivatives.

Moreover, insurers with a high RS ratio also have exposures that are more volatile, suggesting that they frequently rebalance their derivative positions.

4.2.2 Impact of regulatory incentives. (a) *Magnitude of hedging:* Next, I formally examine the impact of the regulatory incentives on the evolution of interest rate risk hedging by estimating the specification below, which relates the shift in interest rate hedging across insurers after 2009, with the cross-sectional variation in the ratio of different types of VA guarantees to total capital,²⁸

$$\widehat{\delta}_{it} = \alpha + \sum_{J=RS, PRS} \beta^J (J_i \times Post_t) + \gamma' X_{it-1} + \alpha_i + \alpha_t + \epsilon_{it}, \quad (15)$$

where $\widehat{\delta}_{it}$ is the total exposure from interest rate derivatives scaled by capital for insurer i at time t . $J=RS$ is the ratio of RS guarantees to capital and $J=PRS$ is the ratio of PRS guarantees to capital, defined in Equation (13). $Post_t$ is a

²⁸ The estimations are at an annual frequency because important control variables (e.g., the RBC ratio, guarantee ratio, and MBS and callable bond share) are only available at an annual frequency. Estimations on quarterly data, without these control variables, produce extremely similar results.

Table 4
Cross-sectional patterns in intensive margin of interest rate risk hedging

	Level $\bar{\delta}_i$	Volatility $\sigma(\hat{\delta}_i)$
$\overline{RS}_i$	1.217*** (0.234)	1.121*** (0.198)
$\overline{PRS}_i$	0.083 (0.074)	0.068 (0.063)
$\overline{\log(assets)}_i$	0.006 (0.005)	0.010** (0.004)
$\overline{Leverage}_i$	0.0018 (0.0015)	0.0013 (0.0013)
$\overline{RBC}_i$	0.0002 (0.0002)	0.0002 (0.0001)
N	53	53
Adj R^2	0.43	0.52

The table reports the results of the cross-sectional linear regression to understand the determinants of the intensive margin of interest rate risk hedging. The dependent variables are (a) time-series average of hedging exposures scaled by capital $\bar{\delta}_i$ and (b) standard deviation of hedging exposures scaled by capital $\sigma(\hat{\delta}_i)$ for each insurer, which I compute using quarterly data from 2004 to 2016. The sample includes insurers that held interest rate derivatives at some point during the sample period. The independent variables are time-series averages of each variable computed for each insurer using the entire available sample. Intercept is estimated, but not reported. The table shows standard errors in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

dummy variable that takes a value of one for years on or after 2009, and X_{it-1} are insurer-specific controls, discussed below. The preperiod is 2004 to 2008 and the postperiod is 2009 to 2016. I include insurer fixed effects, α_i , to control for non-time-varying unobservables across insurers and time fixed effects, α_t , to control for time trends, for example, changes in interest rate conditions that drive hedging demand, which absorb RS_i , PRS_i , and $Post_t$, respectively.

The coefficients of interest are β^{RS} and β^{PRS} , which measure the correlation between the shift in interest rate risk hedging after 2009 and the exposure of insurers to RS and PRS guarantees, respectively. If insurers do not care about regulatory incentives, then interest rate hedging outcomes for both RS and PRS guarantees would be similar as the economic interest rate exposures are similar for both RS and PRS guarantees (see Appendix Table A.2). However, if regulatory incentives matter, then hedging outcomes would be different because regulatory interest rate sensitivities are different for RS and PRS guarantees.

Table 5 presents the results. I estimate a positive and statistically significant coefficient (at the 10% level; column 1) when I use the ratio of total VA guarantees to capital, G_i , as the main regressor. Next, I split the variable into its constituents, RS_i and PRS_i . I estimate a positive and statistically significant coefficient β^{RS} , which captures the interaction between RS_i and the post 2009 dummy, at the 1% level. However, β^{PRS} is statistically insignificant (column 2).²⁹ In columns 3 and 4, I introduce the two variables separately; however, the estimates of β^{RS} and β^{PRS} do not change materially.

²⁹ A test of equality of the two coefficients, β^{RS} and β^{PRS} , is rejected at a confidence level of 1%.

Table 5
Regulatory incentives and magnitude of interest rate risk hedging

	1	2	3	4	5	6	7
$G_i \times Post_t$	0.354* (0.203)						
$RS_i \times Post_t$		1.975*** (0.256)	2.044*** (0.242)		1.955*** (0.235)	1.996*** (0.234)	1.952*** (0.278)
$PRS_i \times Post_t$		0.171 (0.136)		0.238 (0.166)			
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Convexity controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Guarantee ratio $\times$ Year fixed effects					Yes		
Asset convexity $\times$ Year fixed effects						Yes	
Size $\times$ Year fixed effects							Yes
N	624	624	624	624	624	624	624

The table shows the relationship between shifts in interest rate risk hedging and the exposure to different VA guarantees. The dependent variable δ_{it} is the hedging exposure scaled by capital for firm i at time t . The main independent variables are (a) G_i , (b) RS_i , and (c) PRS_i , defined in Equation (13), each interacted with $Post_t$, a dummy variable that takes a value of one for years on or after 2009 and zero before 2009. Estimations are on annual data. The preperiod is 2004 to 2008, and the postperiod is 2009 to 2016. The sample includes insurers that hold interest rate derivatives. The controls include log(assets), leverage, RBC ratio, credit rating, BCAR relative to guideline, and separate account share. The convexity controls are guarantee ratio, and MBS and callable bond share. The table shows standard errors, clustered at the insurer level, in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Figure 5 further corroborates the findings by describing the dynamics of the shift in hedging for the RS and PRS ratios for each year surrounding the introduction of the new regulation relative to a base year (2004). There is no change in hedging in the preperiod, but hedging increases substantially in the postperiod. Moreover, the increase in hedging correlates positively and significantly with RS_i , but not with PRS_i .

The differences in hedging between RS and PRS guarantees are large in magnitude after the regulatory shift. For the *average* insurer that sells VA guarantees, my estimates imply a hedging gap (between RS and PRS guarantees) of 7.1 cents per dollar of capital for a 100-bps shift in interest rates after the regulatory shift, which can be interpreted as a duration gap of roughly 7 years. In other words, capital declines by 7 percentage points less due to a 100-bps decline in interest rates for RS, relative to PRS, guarantees.³⁰

Differential shifts in economic exposures: A natural identification concern is whether economic risk exposures increase faster for high RS insurers than high PRS insurers when interest rates decline due to differences in convexity of exposures. If so, then the RS ratio measures not only the shift in the regulatory risk sensitivities, but also the shift in the economic risk exposures,

³⁰ I multiply 3.9%, which is the PRS ratio of the average insurer that sells VA guarantees, by 1.8, which is the difference in the coefficients β_{RS} and β_{PRS} (Table 5), to get 7.1%. If the average insurer sells only RS guarantees, the shift in the exposures would be 7.8%, whereas it actually is 0.7%, implying a gap of 7.1%.

which could explain the relative increase in interest rate risk hedging over time. However, as Appendix A shows, PRS guarantees have quantitatively similar economic interest rate exposures as RS guarantees under different interest rates and equity market (moneyness) conditions. In other words, the shift in economic exposures, as a result of changes in interest rates and equity markets, is similar for both types of guarantees. Thus, the observed differences in hedging outcomes across guarantees after 2009 implies that insurers indeed care about the regulatory incentives.

To mitigate the concern about differential shifts in economic exposures further, I control for a number of variables that affect balance sheet convexities, which I label as convexity controls in Table 5. On the liability side, more generous guarantees imply a higher convexity because when interest rate are low fewer households surrender their contracts (Domanski, Shin, and Sushko 2017). I control for the generosity of guarantees by the ratio of the value of guarantees to total account value, as in Kojien and Yogo (2022). On the asset side, a higher proportion of mortgage-backed securities (MBS) or callable bonds could also imply a differential shift in exposures because of negative convexity (Hanson 2014), which implies an increase in exposures when rates fall. I control for the share of MBS and callable bonds in total assets. I also include the guarantee ratio quartile and MBS and callable bond share quartile $\times$ year fixed effects in columns 5 and 6. The estimates remain largely unaffected.

I control for a number of characteristics in addition to measures of economic risk exposures, for example, size (log assets), size quartile $\times$ year fixed effects, leverage, RBC ratio, credit rating, and BCAR, as will be discussed in detail below. Collectively, these findings are consistent with **Prediction 1** that the increase in the regulatory risk sensitivities would lead to an increase in hedging with derivatives for RS, but not PRS, guarantees.

(b) *Dynamic hedging*: As risk exposures increase when interest rates fall, an increase in hedging in response to a decline in interest rates reveals dynamic hedging of the underlying exposures. To examine the impact of the regulatory incentives on hedging dynamics, I estimate the specification below, which quantifies the relationship between changes in hedging and changes in interest rates in the cross-section and over time:

$$\Delta \widehat{\delta}_{it} = \alpha + \sum_{J=RS, PRS} \beta_{D1}^J (R_{10t} \times J_i \times Post_t) + \sum_{J=RS, PRS} \beta_{D2}^J (R_{10t} \times J_i) + \sum_{J=RS, PRS} \beta_{D3}^J (J_i \times Post_t) + \gamma' \Delta X_{it} + \alpha_i + \alpha_t + \epsilon_{it},$$

where $\Delta \widehat{\delta}_{it}$ is the quarterly change in hedging and R_{10t} is the quarterly return on a 10-year Treasury bond from CRSP, which captures changes in interest rates. I include insurer fixed effects and time fixed effects, which absorb J_i , $R_{10t} \times Post_t$, R_{10t} , and $Post_t$. The main coefficient of interest is the loading on the triple interaction term β_{D1}^J , which measures the correlation between the

Table 6
Regulatory incentives and hedging dynamics with interest rates

	1	2	3	4	5
R_{10t}	0.133*** (0.05)	0.037 (0.039)			
$R_{10t} \times RS_i$		10.404*** (2.962)	3.849 (2.893)		4.439 (2.859)
$RS_i \times Post_t$			-0.054 (0.044)		-0.047 (0.043)
$R_{10t} \times RS_i \times Post_t$			9.866*** (1.731)		9.671*** (1.693)
$R_{10t} \times PRS_i$		0.423 (0.404)		-0.818 (1.487)	0.567 (1.031)
$PRS_i \times Post_t$				-0.016 (0.013)	-0.015 (0.013)
$R_{10t} \times PRS_i \times Post_t$				0.709 (0.802)	0.407 (0.665)
Controls			Yes	Yes	Yes
Convexity controls			Yes	Yes	Yes
Firm fixed effects		Yes	Yes	Yes	Yes
Time fixed effects			Yes	Yes	Yes
N	2,668	2,668	2,668	2,668	2,668

The table shows the relationship between changes in interest rate risk hedging and bond returns. The dependent variable $\Delta \widehat{\delta}_{it}$ is the quarterly change in hedging for firm i at time t . The main independent variables are the interactions between the quarterly return on a 10-year Treasury bond R_{10t} , RS_i , PRS_i , and $Post_t$, which is a dummy variable that takes a value of one for years on or after 2009 and zero before 2009. RS_i and PRS_i are defined in Equation (13). Estimations are on quarterly data. The preperiod is 2004:Q2 to 2009:Q3, and the postperiod is 2009:Q4 to 2016:Q4. The sample includes insurers that hold interest rate derivatives. The controls are the change in log(assets), change in leverage, size interacted with R_{10t} , guarantee ratio interacted with R_{10t} , and MBS and callable bond share interacted with R_{10t} . The table shows standard errors, clustered at the insurer level, in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

shift in the sensitivity of changes in hedging due to changes in interest rates and RS_i and PRS_i .

Table 6 shows that changes in hedging are sensitive to changes in interest rates (column 1). As interest rates decrease, and bond prices (thus, bond returns) rise, hedging increases. A 1% increase in bond returns leads to a 13.3 (bps) change in hedging exposure per dollar of capital. Moreover, this relationship is positively and significantly correlated with the RS ratio and uncorrelated with the PRS ratio (column 2). As the RS ratio increases, the covariation between changes in hedging and changes in interest rates also increases.

Columns 3-5 show the effects of the shift in the regulation. The coefficient of interest β_{D1}^{RS} is positive and statistically significant (at the 1% level) and economically large in magnitude. Thus, insurers with a high RS ratio have a higher covariation between changes in hedging and interest rates. Moreover, the covariation is significantly higher after 2009 than before 2009. In contrast, I find a negligible relationship between the PRS ratio and the covariation between changes in hedging and interest rates in either sample period. Thus, insurers with a high PRS ratio, not only did not increase hedging, but also did not rebalance old positions as interest rates decreased significantly after 2009,

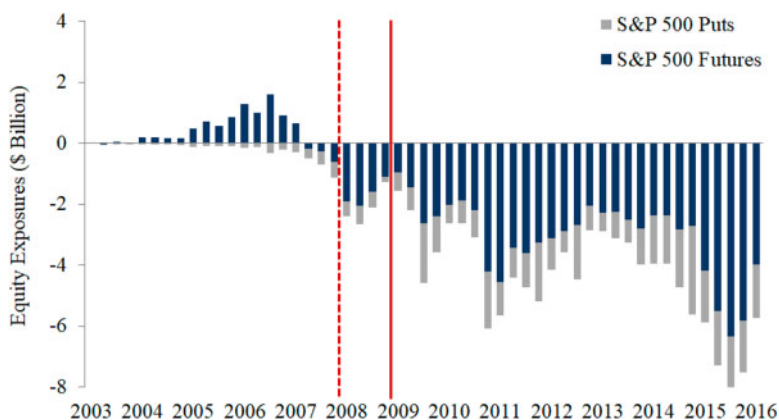


Figure 6
Equity risk exposures from derivatives

The figure shows the total dollar risk exposures from equity derivatives for a 10% shift in the S&P 500 index. I incorporate all contract characteristics, including direction of the position, moneyness, and maturity in computing the risk exposures. The dark bars represent exposures from equity futures, and the light bars represent exposures from put options. Negative (positive) exposures imply that the derivative portfolio gains (loses) in value when the index falls. The solid vertical line (dotted line) represents the implementation (adoption) of the new regulation. Exposures are aggregated for all the insurers in the sample that hold equity derivatives. The data frequency is quarterly from 2004:Q1 to 2016:Q4, and the start date is dictated by the availability of the derivatives data.

which further substantiates the prediction of less hedging amongst insurers that underwrite PRS guarantees.

I control for a number of firm characteristics, for example, changes in size and leverage, size $\times$ bond returns to account for the differential hedging ability of firms, the guarantee ratio $\times$ bond returns, and MBS and callable bond share $\times$ bond returns to account for the differential change in economic exposures of assets or liabilities. The loading on β_{D1}^{RS} remains significant and β_{D1}^{PRS} remains insignificant and similar in magnitude. Overall, the results are consistent with **Prediction 2** that an increase in the regulatory risk sensitivities would lead insurers to dynamically hedge the interest rate mismatch of RS, but not of PRS, guarantees.

4.3 Equity risk hedging

4.3.1 Stylized facts. Next, I discuss the key stylized facts related to the hedging of equity risk of VA guarantees. First, the risk exposures of equity derivatives are *negative*, that is, the equity derivatives gain in value when the stock market declines (Table 3). VA are exposed to a fall in the stock market because the embedded put option becomes in-the-money (and liabilities increase in value) when the stock market declines. A negative exposure of equity derivatives implies that, in turn, insurers have taken long positions in put options and short positions in equity futures, consistent with hedging the equity exposure in a VA guarantee.

Table 7
Regulatory incentives and magnitude of equity risk hedging

	1	2	3	4
$G_i \times Post_t$	-0.191*** (0.062)			
$RS_i \times Post_t$		-0.239*** (0.088)	-0.317*** (0.076)	
$PRS_i \times Post_t$		-0.179** (0.077)		-0.198*** (0.076)
Controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
N	578	578	578	578

The table shows the relationship between shifts in equity risk hedging and the exposure to different VA guarantees. The dependent variable is the equity risk hedging exposure scaled by capital for firm i at time t . The independent variables are (a) G_i , (b) RS_i , and (c) PRS_i , defined in Equation (13), each interacted with $Post_t$, a dummy variable that takes a value of one for years on or after 2009 and zero before 2009. Estimations are on annual data. The preperiod is 2004 to 2008, and the postperiod is 2009 to 2016. The sample includes insurers that hold equity derivatives. The controls include log(assets), leverage, RBC ratio, credit rating, BCAR relative to guideline, and separate account share. The table shows standard errors, clustered at the insurer level, in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Second, Figure 6 shows that just as interest rate exposures increased, equity exposures have also increased substantially since 2009. A 10% shift in the S&P 500 would have resulted in a relatively small change in the value of the equity derivatives portfolio before 2009, however, the same shift would have led to a change of over \$8 billion in 2016 for insurers in the aggregate. For example, there is a sixfold increase in risk exposures from equity derivatives for insurers in aggregate between 2008 and 2016. Similarly, Table 3, panel B shows that the average insurer that holds equity derivatives has had a large increase in equity exposures during the period 2009-2016 relative to the period 2004-2008.

4.3.2 Impact of regulatory incentives. (a) *Magnitude of hedging:* Next, I examine the impact of regulatory incentives on equity risk hedging. The examination of equity hedging outcomes is a useful test for the regulatory incentives hypothesis. Unlike interest rates, the regulatory equity risk sensitivities increased for all guarantees under the new regime as both the Standard and the Stochastic frameworks became more sensitive to shifts in the equity market (Figure 1). Moreover, the economic equity risk exposures are similar across guarantees (Appendix Table A.2). Thus, unlike interest rate hedging, the new regime should result in a similar increase in equity risk hedging for both RS and PRS guarantees. To test whether this is the case, I estimate the model in Equation (15) with equity hedging exposures as the dependent variable.

Table 7 documents the main findings. I estimate a negative and statistically significant coefficient (at the 1% level) for the ratio of total VA guarantees to capital, G_i (column 1). The negative sign simply captures the negative exposures for equity derivatives, that is, long positions in put options and short

positions in futures gain in value when the equity market declines. A larger negative coefficient implies an *increase* in equity risk hedging. Next, I split the variable into its constituents RS_i and PRS_i . Both β^{RS} and β^{PRS} are statistically significant (columns 2-4). Thus, the shift in equity risk hedging after 2009 correlates not only with the RS but also with the PRS ratio.

Figure 5 shows the dynamics of the shift: while there is no change in equity hedging in the preperiod, hedging increases substantially in the postperiod. Moreover, the magnitude of the shift in equity hedging is similar for both the RS and PRS ratios, unlike what I find for interest rate hedging, where the increase in hedging is concentrated only within insurers highly exposed to RS guarantees. My estimates imply a negligible hedging gap (between RS and PRS guarantees) of 0.02 cents per dollar of capital for a 10% shift in the S&P 500 for the average insurer that sells VA guarantees.

(b) *Dynamic hedging*: Insurers hedge the equity risk embedded in VA by dynamically rebalancing their hedging positions. As VA are long-dated options, a static hedging strategy would entail buying equally long-dated put options on the same underlying in the OTC market. Table 1 shows that while insurers tend to buy put options to hedge VA, over 70% of the options have a maturity under 1 year and over 90% are under 5 years. Moreover, Figure 6 shows that 70% of the equity hedging exposures come from short positions in S&P 500 futures, which are 3-month long contracts, and 30% come from put options. Thus, insurers are rolling over short positions in futures and long positions in (mostly 1-year) put options to dynamically rebalance their hedging positions.

Overall, consistent with the differential hedging incentives provided by the regulatory framework, I find that insurers with a high RS ratio increase *both* interest rate and equity risk hedging after the regulatory shift. However, insurers with a high PRS ratio increase equity hedging, but *not* interest rate risk hedging.

4.4 Regulatory constraints and alternative explanations

4.4.1 Regulatory constraints. The model shows that an increase in the regulatory risk sensitivities would generate a larger increase in hedging for insurers that are constrained relative to those that are unconstrained prior to the regulatory changes (see Prediction 3). To evaluate this relationship between shift in hedging and regulatory constraints, I divide insurers into two groups by their RBC ratio in 2007 and reestimate Equation (15) for each group separately. Low (below-median) RBC are insurers that are more likely to be constrained and high (above-median) RBC are insurers that are more likely to be unconstrained. Table 8 shows that the impact of the shift in the regulation on both interest rate and equity hedging is more pronounced for insurers that are constrained, consistent with **Prediction 3(A)**.

Table 8
Hedging and regulatory constraints: Cross-sectional evidence

	Interest rate risk		Equity risk	
	Low RBC	High RBC	Low RBC	High RBC
$RS_i \times Post_t$	2.126*** (0.095)	1.03 (0.712)	-0.334*** (0.045)	-0.229 (0.357)
$PRS_i \times Post_t$	0.176 (0.126)	-0.218 (0.339)	-0.149* (0.081)	0.357* (0.209)
Controls	Yes	Yes	Yes	Yes
Convexity controls	Yes	Yes		
Firm fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
N	318	306	282	296

The table shows the estimations on two equal subgroups: low RBC and high RBC, using the RBC ratios in 2007. The first two columns show results on interest rate risk hedging and the next two columns show results on equity risk hedging. The independent variables are RS_i and PRS_i , defined in Equation (13), each interacted with $Post_t$, a dummy variable that takes a value of one for years on or after 2009 and zero before 2009. Estimations are on annual data. The preperiod is 2004 to 2008, and the postperiod is 2009 to 2016. The sample includes insurers that hold interest rate or equity derivatives. The controls include log(assets), leverage, RBC ratio, credit rating, BCAR relative to guideline, and separate account share. The convexity controls are guarantee ratio, and MBS and callable bond share. The table shows standard errors, clustered at the insurer level, in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

To examine how hedging covaries with RBC and how this relationship varies in the cross-section and over time, I estimate the following model:

$$\begin{aligned} \widehat{\delta}_{it} = & \alpha + \beta_{R1}^{RS}(RBC_{it-1} \times RS_i \times Post_t) \\ & + \beta_{R2}^{RS}(RBC_{it-1} \times RS_i) + \beta_{R3}^{RS}(RBC_{it-1} \times Post_t) \\ & + \beta_{R4}^{RS}(RBC_{it-1}) + \beta_{R5}^{RS}(RS_i \times Post_t) + \gamma' X_{it-1} + \alpha_i + \alpha_t + \epsilon_{it}, \end{aligned}$$

where I measure RBC at time $t - 1$. All the remaining variables are defined in Equation (15). β_{R1}^{RS} measures the shift in the covariation after 2009 due to a shift in the regulatory risk sensitivities. β_{R2}^{RS} measures the baseline covariation at different RS ratio levels. β_{R3}^{RS} and β_{R4}^{RS} measure the shift in the covariation after 2009 and the baseline covariation for insurers with no RS guarantees, and therefore no regulatory interest rate sensitivity.

Appendix Table C.2, column 1, shows that hedging the RBC ratio are positively (and statistically significantly) related. As the RBC ratio falls (constraints become tighter), hedging decreases. Moreover, the covariation between RBC and hedging changes over time and varies in the cross-section, depending on the RS ratio (columns 2-3). Low RS insurers (low regulatory sensitivity) hedge less when RBC declines. In contrast, high RS insurers (high regulatory sensitivity) hedge more when RBC declines, but only after 2009 when regulation shifts. The estimates of the full model in column 4 confirm these findings. β_{R1}^{RS} is negative and significant and β_{R2}^{RS} is positive and significant. Thus, the covariation between hedging and RBC changes sign after 2009, that is, insurers start to hedge more as they become more constrained. Moreover, the shift in the covariation is stronger for insurers who

experienced a larger shift in the regulatory risk sensitivities. In addition, β_{R4}^{RS} is positive and statistically significant and β_{R3}^{RS} is insignificant. Thus, insurers with no regulatory interest rate risk sensitivity ($RS_i=0$), hedge less as they become more constrained, and there is no shift in the covariation over time. Thus, consistent with **Prediction 3(B)**, regulatory constraints affect hedging in opposite ways depending on the risk sensitivity of the framework.

4.4.2 Alternative explanations. My setting and empirical specifications address several alternative explanations. I first examine the extent to which the measures of the shift in regulatory sensitivities (RS_i and PRS_i) correlate with other firm characteristics that may also affect hedging. Appendix Table C.3 shows that the RS and PRS ratios do not correlate with most characteristics, except insurers' size. This is not surprising because my analysis conditions on insurers which hold derivatives and are, therefore, likely to be highly similar along most dimensions. However, differences in size could proxy for an insurer's sophistication and ability to bear the fixed costs of hedging, which I address below along with other alternative explanations.

(a) *Sophistication and fixed hedging costs.* A number of tests help to rule out this explanation. First, all insurers hedge equity, but only a subset hedge interest rate risk. Sophistication and fixed hedging costs are unlikely to explain why some insurers hedge one risk, but not the other. Second, I examine how much insurers hedge (intensive margin) and not whether they hedge (extensive margin) as the estimations occur within an insurer controlling for insurer fixed effects. Moreover, my sample conditions on insurers that hold derivatives; thus, there is likely limited heterogeneity in firm's sophistication across my sample of insurers. Third, I include the log of total assets (size) in all the specifications. I also include size quartile $\times$ year fixed effects. The estimated coefficients are robust to these control variables.

(b) *Shifts in risk aversion and exposure to the financial crisis.* Heterogeneity in risk aversions could lead to different hedging choices, even though the underlying economic exposures are the same. All specifications use insurer fixed effects, which help account for static differences in risk aversion across firms. However, risk aversions could have possibly shifted for a subset of insurers who were differently exposed to the financial crisis. In addition, it is also possible that insurers learned about the risks of VAs during the financial crisis, and this prompted more hedging. However, the results are unlikely to be driven by the differential exposure to (and learning from) the financial crisis.

First, if increase in risk aversion is driving the results, then it is unclear why insurers exposed to PRS guarantees do not increase interest rate hedging, but only increase equity hedging. Second, the empirical specifications control for the RBC ratio, leverage, and BCAR relative to the guideline, which proxy for financial constraints and the exposure to the financial crisis. Third, to explicitly account for the differential exposure to the financial crisis, I include the size of the shock experienced during the crisis, measured as changes in

capital between 2007:Q4 and 2009:Q1, interacted with a postcrisis dummy in Appendix Table C.4 ($Shock_i \times Post_t$). If the changes in hedging are driven just by the increase in risk aversion of a subgroup of insurers that suffered a large shock during the crisis, then $Shock_i \times Post_t$ should be statistically and economically significant and the loadings on RS and PRS ratios should be subsumed. However, the loadings on the RS and PRS ratios remain unaffected and $Shock_i \times Post_t$ is insignificant, which imply that the increase in hedging is not just concentrated among insurers that were adversely affected during the crisis.

(c) *Collateral constraints.* A prominent explanation for the absence of hedging is collateral constraints (Rampini and Viswanathan 2010). In these models, firms' future investments and risk management both compete for scarce capital. The key state variable is firms' economic net worth and the prediction is that low net worth firms, because they have a greater opportunity cost, engage in less risk management. However, my central result cannot be explained by changing net worth of a certain set of insurers. It is unlikely that the regulatory change that I study perfectly maps to a certain subgroup of insurers which had their collateral constraints loosened in 2009. Having said that, the regressions control for the size of the shock experienced during the crisis, guarantee value, leverage, and RBC ratio, all of which proxy for financial constraints. Moreover, net worth constraints operate at the firm level. But I find heterogeneity in which risk exposures firms hedge and which they do not. These choices line up with the incentives provided by regulation.

Overall, the observed hedging behavior is consistent with the differential incentives provided by the regulatory framework, and not with cross-insurer differences in sophistication, fixed hedging costs, risk aversion, exposure to the crisis, or financial constraints.

5. Alternative Hedging Strategies

In this section, I evaluate whether insurers hedge in other ways and regulation simply pushes them from one type of hedging to another. I evaluate two alternative hedging strategies: reinsurance and securities (bonds), which as discussed in Section 1, are other possible ways to hedge VA guarantees.

5.1 Reinsurance

Using reinsurance, insurers can transfer risk exposures off-balance sheet, either to third-party reinsurers or to captives (also known as shadow insurers, which are special purpose vehicles and offshore domiciled entities within the same holding company). I evaluate two hypotheses related to the use of reinsurance as a risk management tool for VA guarantees. *Hypothesis 1:* Insurers shift risk management from reinsurance to derivatives after the regulation changes, but overall hedging remains the same. *Hypothesis 2:* Insurers highly exposed to PRS guarantees (that do not hedge interest rate risk using derivatives) should

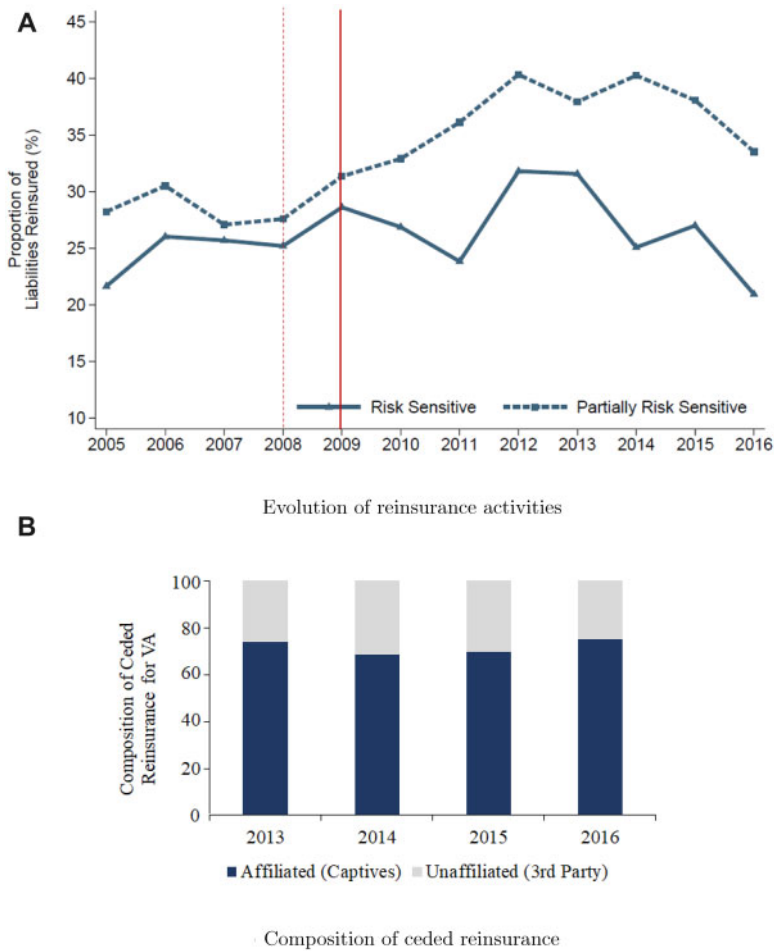


Figure 7

Regulatory incentives and off-balance-sheet risk transfers

Panel A shows the evolution of proportion of liabilities reinsured for the two types of VA guarantees: RS and PRS. The solid vertical line (dotted line) represents the implementation (adoption) of the new regulation. The sample comprises insurers that hold interest rate derivatives, and the data frequency is annual from 2005 to 2016. Panel B shows the breakdown of reinsurance for VA into captives (shadow insurers) and third-party reinsurers at the aggregate level. The start dates are dictated by data availability.

engage in more reinsurance if they want to hedge PRS risks using other ways. To evaluate these hypotheses, I exploit data on reinsurance at the guarantee level, which insurers report at an annual frequency in the statutory filings (see Section 3).

5.1.1 Is there substitution from reinsurance to derivatives? To evaluate whether there is substitution from reinsurance to derivatives after the regulatory

Table 9
Reinsurance and off-balance-sheet transfers

<i>A. Shift in reinsurance</i>			
	PRS	RS	Difference
$Post_t$	0.077** (0.039)	-0.005 (0.026)	0.082* (0.043)
Controls	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
N	510	510	510
<i>B. Impact of regulatory incentives</i>			
	PRS		
$RS_i \times Post_t$	-1.254 (2.079)		
$PRS_i \times Post_t$	1.63*** (0.299)		
Controls	Yes		
Firm fixed effects	Yes		
Time fixed effects	Yes		
N	510		

Panel A shows the estimations to quantify the shift in reinsurance for RS and PRS guarantees. The dependent variables are the proportion of RS liabilities reinsured, PRS liabilities reinsured, and the difference between proportion of PRS and RS liabilities reinsured for firm i at time t . The main independent variable is a dummy variable $Post_t$ that takes a value of one for years on or after 2009. The controls are log(assets), leverage, RBC ratio, credit rating, and BCAR relative to guideline. Panel B shows the relationship between shift in reinsurance and the shift in regulatory incentives. The dependent variable is the proportion of PRS liabilities reinsured. The independent variables are RS_i and PRS_i , defined in Equation (13), each interacted with $Post_t$. Estimations are on annual data. The preperiod is 2005 to 2008, and the postperiod is 2009 to 2016. The table shows standard errors, clustered at the insurer level, in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

changes, I ask whether reinsurance declines after 2009, particularly for RS guarantees for which hedging with derivatives increased substantially. Figure 7 plots the proportion of liabilities reinsured for the two types of guarantees over time. Before 2009, on average about 25% to 30% of liabilities were reinsured for both guarantees. After 2009, there is no evidence to suggest that reinsurance declined for either RS or PRS guarantees. To formally evaluate the shifts in reinsurance behavior, I regress the proportion of liabilities reinsured on a post-2009 dummy variable. Table 9 (panel A) documents no meaningful reduction in the use of reinsurance for either RS or PRS guarantees after 2009, suggesting that insurers do not substitute out of reinsurance and into derivatives.

5.1.2 Do insurers hedge PRS risks with reinsurance more? Figure 7 shows that relative to RS guarantees, a greater proportion of PRS guarantees are reinsured after the regulation shifts. Moreover, Table 9 (panel A) documents that the average insurer increased reinsurance by 8 percentage points more for PRS guarantees than for RS guarantees after 2009. This constitutes an economically large (29%) increase in PRS reinsurance for the average insurer.

Regulatory incentives. As reinsurance behavior also could be potentially driven by non-hedging motives, for example, capital and tax management (Tiller and Tiller 2009), it is unclear whether the shift in reinsurance is solely

due to changes in the regulatory risk management incentives. To confirm the regulatory incentives hypothesis, I reestimate Equation (15) with the proportion of PRS liabilities reinsured as the dependent variable. Table 9 (panel B) shows that the increase in PRS reinsurance after 2009 is correlated with the ex ante PRS ratio, that is, the increase is concentrated among insurers who do not hedge interest rate risk exposures with derivatives.

As changes in reinsurance activities also have been linked to financial constraints (Kojen and Yogo 2016), these regressions control for insurers' financial position using a number of variables, including leverage, RBC ratio, credit rating, and BCAR ratio. I also control for insurers' size using $\log(\text{assets})$ to account for different abilities to engage in reinsurance. Overall, these findings show that insurers want to hedge PRS risks using ways other than derivatives. As a result, the composition of risk management for RS and PRS guarantees undergoes a shift after the regulation changed. RS exposures are increasingly hedged with derivatives, while PRS exposures are moved off-balance sheet using reinsurance.

5.1.3 Why do insurers not use even more reinsurance for PRS guarantees?

Two potential reasons could explain why insurers do not use even more reinsurance for PRS guarantees. First, Figure 7 (panel B) shows that close to 75% of all reinsurance for VA goes to captives (shadow insurers), which are less regulated and often deemed as potential regulatory blind spots (OFR 2016). As a result, state regulators place limits on how much shadow insurance is permitted (Lawsky 2013). Second, third-party reinsurance can be costly (Froot and O'Connell 1999). These factors can limit the amount of PRS exposures that can be moved off-balance sheet using reinsurance.

5.2 Hedging with bonds

An alternative way to hedge the interest rate risk of VA guarantees is to buy a matching portfolio of bonds. To understand to what extent insurers hedge with bonds, I evaluate the following two hypotheses. *Hypothesis 3*: Insurers are already well hedged with bonds, which is why they do not hedge with derivatives prior to the shift in the regulation. *Hypothesis 4*: Insurers shift risk management from bonds to derivatives after the regulation changes but overall hedging remains the same.

Data and measurement: To evaluate these hypotheses, I exploit detailed holdings of all fixed income securities, including treasuries and government bonds, municipal bonds, agency and mortgage backed bonds, and corporate bonds, which insurers report at a CUSIP level in NAIC's schedule D filings. The data are annual and span the period from 2004 to 2016. For each position, I observe the total par and market values, remaining maturity, coupon paid, and rating. To quantify the magnitude of interest rate risk hedging from bonds, one must account for both the size of the bond portfolio and the bonds' maturities. To do so, I measure the interest rate exposures of bond holdings

(bond exposures). Specifically, I compute the total dollar duration of bond holdings, δ_{it}^B , for each insurer i at the end of each year t .³¹ Dollar duration quantifies the total dollar change in the value of the bond holdings due to a 100-bps parallel shift in the yield curve.

5.2.1 Are insurers hedging VA guarantees with bonds? To assess to what extent insurers use bonds to hedge the interest rate risk in VA guarantees, I regress the total dollar duration of bond holdings (bond exposures) on the various types of insurance liabilities:

$$\widehat{\delta}_{it}^B = \beta_1 G_{it} + \beta_2 Life_{it} + \beta_3 FA_{it} + \beta_4 IA_{it} + \gamma X_{it} + \alpha_t + \alpha_i + \epsilon_{it},$$

where $\widehat{\delta}_{it}^B$ is the total dollar duration of bond holdings scaled by total capital for firm i at time t . Independent variables G_{it} , $Life_{it}$, FA_{it} , and IA_{it} denote VA guarantees, life, fixed annuities, and indexed annuities liabilities scaled by total capital, as described in Section 3.2. The sample includes *all* insurers, including those that sell VA and those that do not.

Appendix Table C.5 shows that the relationship between bond exposures and VA guarantees is not just statistically insignificant but also economically less important. To summarize the main findings, I find that (a) in the cross-section (column 1), higher bond exposures are associated with a higher Life, FA, and IA liabilities (β_2 to β_4 are positive and significant at the 1% level), but *not* higher VA guarantees (β_1 is insignificant). (b) The within-insurer variation (with insurer fixed effects) in bond exposures are also *only* related to non-VA liabilities (column 2). Thus, bond exposures comove with life, FA, and IA liabilities over time, but not with VA guarantees. The interpretation is that a dollar of life liabilities adds roughly 10 cents of dollar duration to the average insurer's portfolio. In contrast a dollar of guarantees adds only 1.2 cents of dollar duration. (c) In columns 3 and 4, I exclusively focus on the preregulation period³² and confirm that the same patterns hold true; that is, the cross-sectional and the within-insurer variation in bond exposures are related to non-VA liabilities *only* and not with VA guarantees.³³ Overall, these findings show that the evidence is not consistent with the hypothesis that insurers hedge VA guarantees with bonds in a meaningful way.

³¹ Total dollar duration is given by $\delta_{it}^B = \sum_j Dur_{ijt} \times B_{ijt}$, where Dur_{ijt} is a bond j 's Macaulay duration and B_{ijt} is its market value. I exclude positions where market value or maturity date is missing. I also exclude floating rate positions, which have small interest rate exposures. I treat all bonds as straight bonds.

³² A breakdown of annuities into its components (IA and FA) is unavailable prior to 2010, so I use the total non-VA annuities as the regressor.

³³ Why do the regulatory incentives end up being different for non-VA liabilities relative to VAs? Regulatory incentives are determined by how both the liability and its hedges are treated. Regulation treats both liabilities and bonds as risk insensitive. Therefore, the type of regulatory incentives I focus on do not affect non-VA. VAs, on the other hand, require insurers to hedge using derivatives not bonds (see below) and it is difficult to get hedge accounting for derivatives (see Appendix A). As a result, derivatives are marked-to-market, while VA liabilities are not thereby affecting hedging incentives of VAs.

Next, I document the highly similar hedging patterns (with bonds) for insurers with and without VA guarantees. Motivated by the fact that bond exposures are primarily related to non-VA liabilities, I compare the bond exposures per unit of non-VA liabilities for insurers with and without VA guarantees, that is, for insurers *with* VA, I only scale bond exposures by the non-VA portion of their liabilities. Appendix Table C.6 shows that for every dollar of non-VA liabilities, both the groups on average have roughly 10 cents of dollar duration from bonds. A test of difference in means reveals no significant differences, implying that both the groups hedge non-VA liabilities in a similar way with bonds. Moreover, bond exposures (per unit of non-VA liabilities) are also highly stable for both groups over time. This suggests that insurers adjust bond exposures as non-VA liabilities change, consistent with the evidence in Appendix Table C.5 that insurers hedge non-VA liabilities with bonds.³⁴

5.2.2 Is there substitution from bonds to derivatives? Next, I evaluate whether insurers shift out of bonds and into derivatives once the regulation changes. To do so, I examine how bond exposures have changed after the shift in the regulation for insurers with different exposures to RS and PRS guarantees by estimating the specification in Equation (15) with the dependent variable as bond exposures, $\widehat{\delta}_{it}^B$. If insurers exposed to RS guarantees, for whom hedging with derivatives increased substantially, substituted from bonds and into derivatives once the regulation changed then we expect β^{RS} to be negative. Moreover, if overall hedging remained the same, we expect no change in the *total hedging* exposures, which include both derivatives and bonds.

Appendix Table C.7 negates these predictions. First, columns 1-3 show estimations on bond exposures only. β^{RS} is statistically indistinguishable from zero. In particular, there is no evidence of a decrease in bond exposures after the regulation changes for high RS insurers. Second, columns 4-6 show estimations on bond and derivatives exposures jointly. β^{RS} is positive and significant; that is, there is an increase in *total hedging* for high RS insurers and the increase in hedging with derivatives is not cancelled out by a simultaneous decrease in hedging with bonds. Thus, the evidence is not consistent with the hypothesis that insurers substitute out of bonds and into derivatives and that overall hedging remains the same. Moreover, there is no meaningful change in hedging for high PRS insurers with bonds (β^{PRS} is insignificant in column 1). Combined with the result that bond exposures are not correlated with VA guarantees, this implies that insurers highly exposed to PRS guarantees do not hedge with bonds despite the lack of hedging with derivatives.

5.2.3 Why do insurers not hedge VA guarantees with bonds? Three reasons could potentially explain why insurers do not hedge VA guarantees

³⁴ This suggests that there is decentralization of risk management activities, consistent with similar trends in asset management (Blake et al. 2013).

with bonds.³⁵ First, VA guarantees have complex payoffs and risk exposures. As a result, they are easier to hedge using derivatives than bonds.³⁶ Second, hedging VA guarantees requires frequent rebalancing³⁷ (also see Table 6), and it may be more difficult to rebalance bonds without price impact. Indeed, insurers' bond holdings are mostly corporate bonds and tend to be illiquid. Thus, trading gives rise to large price impact (Bretscher et al. 2020; Ellul et al. 2020; Sen and Sharma 2020). Moreover, gains and losses from trading in bonds affect capital in the same way as MTM changes in the value of derivatives. Third, because of the cash flow structure of VAs, swaps are preferred because they provide the same exposure as bonds but require limited upfront payment.³⁸

6. Quantifying and Interpreting the Risk Management Shifts

Overall, this paper shows that insurers hedge the specific risks that the regulatory regime starts recognizing. An immediate implication of my findings is that because the regulatory regime does not yet recognize interest rate risk for PRS guarantees, interest rate mismatch still remains. In this section, I quantify the economic significance of the regulatory changes and their impact on insurers' hedging and risk exposures. I also discuss interpretations of the risk management shifts and how the new regime interacts with interest rates in a way that can further exacerbate risk mismatch going forward.

6.1 Quantifying overall hedging and insurers' interest rate exposures

I start by quantifying hedging for the insurance sector in the aggregate incorporating all risk management tools. To determine the fraction of VA exposures hedged, I need three main inputs for each guarantee: (1) estimates of gross risk exposures before hedging, as risk exposures are not directly reported; (2) the fraction of liabilities hedged using alternative ways (reinsurance). I ignore bonds because insurers do not hedge VA with bonds (see Section 5.2); and (3) estimates of offsetting exposures from derivatives due to the shift in the regulation. Table 10 documents the key ingredients.

³⁵ This subsection is developed based on a telephone interview with Ken Mungan (Chairman) and Peter Sun (Principal and Consulting Actuary) at Milliman (August 4, 2021). I refer to this interview as Milliman (2021) throughout the paper.

³⁶ That investors prefer derivatives to hedge more complex risks is true in other contexts. For example, U.S. investors exposed to interest rate risks in mortgage-backed securities have largely relied on interest rate swaps rather than treasuries for dynamic hedging (Perli and Sack 2003; Reinhart and Sack (2000)).

³⁷ Underlying risk exposures change frequently with changes in both interest rates and equity market. When equity markets do well (and guarantees become out-of-the-money), interest rate exposures would be small. A sudden decline in the equity market, however, could give rise to a large increase in exposures.

³⁸ Swaps are levered positions on the underlying bond. The MTM value of swaps is lower than that of bonds (of similar exposure) as the fixed and the floating legs offset each other. Klingler and Sundaresan (2019) argue that underfunded pension funds use swaps instead of bonds to match long-dated liabilities. Domanski, Shin, and Sushko (2017) make the same argument in the context of guaranteed insurance liabilities in Europe.

Table 10
Quantifying hedging

	RS	PRS	Total
<i>A. Account value (\$ billion)</i>			
Pre	250	948	1198
Post	476	1019	1495
<i>B. Gross exposures before hedging (\$ billion)</i>			
Pre	8.2	28.5	36.7 (15%)
Post	23.0	43.1	66.1 (22%)
<i>C. Fraction reinsured (%)</i>			
Pre	25	28	27
Post	30	40	37
<i>D. Net exposures after reinsurance (\$ billion)</i>			
Pre	6.1	20.5	26.7
Post	16.1	25.9	42.0
<i>E. Exposures from derivatives (\$ billion)</i>			
Pre	0	0	0
Post	12.4	1.2	13.6
<i>F. Fraction hedged with derivatives (%)</i>			
Pre	0	0	0
Post	77	5	32
<i>G. Total hedging (%)</i>			
Pre	25	28	27
Post	84	43	57

The table documents the different components of hedging for insurers in the aggregate. Panels A and B report the account values and the gross exposures due to a 100-bps change in interest rates, respectively, where the exposures are computed using the pricing model in Appendix A. Numbers in parentheses are gross exposures as a proportion of total capital. Panels C and D report the average fraction of liabilities reinsured and the exposures net of reinsurance. Panel E reports the offsetting exposure from derivatives, computed using the estimation described in Section 4. Panel F reports fraction of net exposures hedged by derivatives. Panel G reports total hedging from both derivatives and reinsurance. The preperiod is 2007, and the postperiod is 2012. I pick 2007 and 2012 for ease of comparison because the 2 years had similar equity market levels (S&P 500 $\approx$ 1,400). I calibrate 2007 to an interest rate of 4% and 2012 to 2%, consistent with the prevailing 10-year yield during those years to compute the gross exposures of the guarantees.

To estimate the risk exposures of the guarantees, I proceed as follows. First, I value each type of guarantee and compute the risk sensitivities numerically using a standard pricing model (see Appendix A). Specifically, I calculate the ratio of the guarantee value V to the account value F . Second, I obtain the total estimated value of the guarantees by multiplying the estimated V/F with the actual account values, which insurers report in statutory filings.³⁹ Third, to obtain exposures, I multiply the total estimated value of each guarantee by the risk sensitivities, which measure the exposures per dollar of V due to a 100-bps change in interest rates. For each guarantee, I assume average contract features

³⁹ I am assuming that the account value and the notional value are the same, which would be true if the guarantees are at-the-money. In robustness, I allow the contract's moneyness to vary to check the sensitivity of my results to this assumption.

most commonly observed in the market. In my baseline simulations, I assume that the guarantees are at-the-money and in robustness, I vary the guarantees' moneyness.

Panels A and B of Table 10 report the account values and the gross exposures for years 2007 and 2012. I pick 2007 and 2012 for ease of comparison because the 2 years had similar equity market levels (S&P 500 $\approx$ 1,400). I calibrate 2007 to an interest rate of 4% and 2012 to 2%, consistent with the prevailing 10-year yield during those years. For a realistic range of parameter specifications, my pricing model confirms that insurers have substantial interest rate exposures from VA guarantees prior to the regulatory change. In aggregate, gross exposures equaled \$37 billion, or 15% of total capital, due to a 100-bps shift in interest rates in 2007. In the postperiod, exposures go up, both because of an increase in the amount of VAs being written and because of the decline in interest rates.

Panel C reports the fraction of liabilities reinsured on average, and panel D reports the exposures of guarantees net of reinsurance, where I assume that insurers get full hedging from reinsurance. Roughly 27% of the liabilities are reinsured in 2007. Thus, net of reinsurance there still remains substantial interest rate mismatch prior to the regulatory shift.

To compute the exposures from derivatives due to the regulatory shift, I exploit the estimates β_{RS}^t and β_{PRS}^t in Figure 5, which measure the average *change* in derivatives exposures in year t relative to a base year per unit of capital for insurers with a certain amount of RS and PRS guarantees relative to insurers with no guarantees. To get derivatives exposures in year t , I assume that insurers have negligible exposures in the base year, consistent with data. For example, Figure 4 shows negligible interest rate exposures in the base year (2004). I also assume that insurers that have no exposure to RS and PRS guarantees have no interest rate derivatives exposures. This is also consistent with data. Insurers with no VA guarantees hold negligible quantities of derivatives (see Figure 3). I then multiply the estimated β_{RS}^t and β_{PRS}^t with the RS and PRS ratio for each insurer and obtain dollar exposures by multiplying by total capital. Panel E shows that there is minimal offsetting exposure from derivatives for both RS and PRS guarantees in 2007.⁴⁰ However, exposures from derivatives increase substantially for RS guarantees in 2012.

6.1.1 Interest rate hedging and risk exposures. To quantify the magnitude of hedging, I compute the ratio of exposures from derivatives (panel E) to the exposures of the guarantees net of reinsurance (panel D). Derivatives hedge close to 77% of the net interest rate exposures for RS guarantees, compared to only 5% for PRS, after the regulation shifts (panel F). Thus, while the regulation meaningfully impacts exposures of insurers with RS guarantees,

⁴⁰ This is because of negligible exposures in the base year (2004), as Figure 4 shows, and the coefficients in Figure 5 for 2007 are insignificant.

insurers with PRS guarantees still have a large mismatch. Finally, panel G reports total hedging, which is the ratio of offsetting exposures (from both derivatives and reinsurance) to gross exposures. After the regulatory changes, the hedging gap between RS and PRS increase substantially: insurers hedge over 80% of RS exposures, the bulk of which takes place through derivatives. However, a large fraction of PRS exposures ($\sim 60\%$) remain unhedged.

More broadly, the remaining interest rate exposures of VA guarantees (after accounting for reinsurance and derivatives) is substantial. For example, in 2012, the exposures equal \$28.4B (9.5% of total capital). This suggests that a 100-bps decline in rates would lead to a 9.5% decline in regulatory capital for the insurance sector in the aggregate. Moreover, bulk of the exposures arise from PRS guarantees. This is because PRS constitute a large share (two-thirds) of VA liabilities (panel A) and the hedging gap between RS and PRS is large.

These estimates are consistent with the large sensitivities of insurers' stock returns to interest rate changes. Appendix Table C.9 measures the sensitivities of stock returns to the equity-market and 10-year Treasury returns. The loading on 10-year Treasury returns is high for large VA underwriters. For example, insurers in the high (above-median) VA group have a sensitivity of -1.2 . This means that the stock returns' sensitivity of large VA underwriters is close to the sensitivity of a 10-year bond, which has a duration of about 9.1 years. A coefficient of -1.2 , thus, can be interpreted as a duration of ~ 11 years (1.2×9.1). In other words, a 100-bps decline in rates would cause the average insurer's equity in this group to decrease by 11%, which is similar in magnitude to the estimates obtained in Table 10.⁴¹ Appendix D discusses additional robustness for the estimates presented in this section.

6.2 Interpreting the risk management shifts

One could interpret the empirical results in two possible ways depending on insurers' baseline hedging motives.⁴² One interpretation is that prior to the regulatory changes insurers wanted to hedge more, but regulation disincentivized hedging by not properly recognizing risks (Kojen and Yogo 2022; Credit Suisse 2012). Prior to 2009, liabilities were largely insensitive, while derivatives were marked-to-market. As a result, if insurers had tried to properly hedge their economic risks under the old regulatory regime, that would have generally increased the volatility of their regulatory capital. To examine whether this is the case, I recompute insurers' regulatory capital (prior to 2009) assuming their derivatives usage was more commensurate with their

⁴¹ It is worth noting that stock returns' sensitivities may not reflect the underlying risk exposures. (a) There may be limited disclosure of risk exposures. (b) Parent companies' sensitivities are also affected by the interest rate exposures of other subsidiaries. (c) Sensitivities likely also capture investors' perceptions about the sensitivity of future demand for insurance to interest rates and not just current exposures.

⁴² It is not necessary to know the the baseline hedging motives for the purpose of identifying the effects of the regulatory changes – under either interpretation, the design of the regulatory framework affects insurers' hedging decisions. However, it is useful for interpreting the empirical facts.

economic exposures. Appendix Figure C.1 shows that had insurers hedged economic exposures prior to 2009 in the same way as they did after the regulatory changes, the volatility of regulatory capital *with* derivatives would have been substantially higher than the volatility of regulatory capital *without* derivatives.⁴³ Since the volatility of regulatory capital can be costly, it can then create incentives for insurers to not hedge.

An alternative interpretation is that insurers did not want to hedge, for example, because of risk shifting motives due to the existence of government guarantees (Lee, Mayers, and Smith 1997). The new regime - by recognizing risks - simply forced insurers to hedge some VA guarantees more. The evidence on reinsurance, discussed in Section 5, however, is consistent with the view that insurers want to hedge their economic risks. We observe insurers employing more reinsurance for PRS relative to RS after 2009, which suggests that they want to hedge PRS even though the regulatory regime does not fully incentivize hedging of PRS risks. This suggests that the introduction of the new regime helped restore insurers' hedging incentives, but only for a subset of VA liabilities as the regulation does not yet recognize interest rate risk for PRS guarantees.

6.3 Monetary policy and financial fragility

An important aspect of the new regime is that it can stop recognizing interest rate risk altogether when rates increase. When interest rates are low, the Stochastic framework dominates with a higher likelihood because the value of the embedded guarantees rise. As interest rates increase, the value of Stochastic framework liabilities reduce and could fall well below Standard framework liabilities, which are fixed and do not change as interest rates change. Using surveys, Junus and Motiwalla (2010) estimate that the Standard framework dominated for a large number of VA underwriters when the regulation was first introduced. As interest rates fell, the Stochastic framework started to dominate. Thus, going forward, as interest rates start to rise, the Standard framework may start to dominate once again and the incentives to hedge could switch off even for guarantees that are currently risk sensitive.

In Appendix Table C.10, I find suggestive evidence that the regulatory framework interacts with interest rates to affect hedging incentives. The threshold rate above which the Standard framework binds varies in the cross-section. This is because the valuation liabilities under the Standard framework depends on the valuation rate (the prevailing rate at the time a contract is

⁴³ Note that the relevant metric is not the volatility of regulatory capital with the actual amount of derivatives insurers used. But it is the volatility of regulatory capital in a (counterfactual) world where the amount of derivatives insurers used would properly hedge the economic risks. This is because in the old regulatory regime, the risks of liabilities were recognized in a few states of the world. For example, to some extent, equity market movements were recognized during extreme downturns (see Section 1). As a result, in such instances some amount of derivatives could have been possibly used, and these derivatives would then reduce the volatility of capital (e.g., in 2008). In fact, Appendix Figure C.1 confirms that insurers are more likely to use derivatives when doing so did not increase the volatility of regulatory capital.

originated). If insurers underwrote when rates were low (high), they would have a lower (higher) valuation rate and a higher (lower) Standard framework liabilities. This would imply a higher (lower) propensity to have the Standard framework binding. Using sales data from Morningstar Annuity to identify insurers for whom the Standard framework is more likely to bind, in Appendix Table C.10, I find that insurers that have underwritten in years when rates were low (and thus have a higher likelihood of the Standard framework binding) have a greater likelihood to shut down hedging when rates rise. This implies that when interest rates eventually rise, incentives to hedge could switch off completely. This can further exacerbate insurers' risk mismatch going forward.

7. Conclusion

Despite large and rapid shifts in risk regulation since the financial crisis, little systematic evidence exists on the ways in which hedging decisions of financial institutions are shaped by the regulatory framework they operate under. In this paper, I quantify the hedging incentives of U.S. life insurers who sell investment products containing long-dated minimum return guarantees called Variable Annuities. I measure the risk exposures of insurers' derivatives portfolio over a long period of time and across risk factors. I show that VA guarantees with similar economic risks are treated differently by regulation and these differences impact insurers' hedging behavior. When the regulatory regime recognizes certain risks, insurers start to hedge these risks in a substantial way. For some guarantees, this involves hedging both interest rate and equity risks, but, for others, it involves hedging only equity risk. Because the regulatory regime does not yet recognize interest rate risk for several VA guarantees, insurers remain exposed to substantial interest rate risk. My findings have implications for the fragility of the insurance sector, which is exacerbated by shifts in interest rates as regulatory incentives to hedge shut down when interest rates rise.

There are several reasons to regulate life insurers. Regulation aims to mitigate the moral hazard and the excessive risk-taking that arises because insurers finance themselves too cheaply due to the presence of state guaranty funds (Bohn and Hall 1999) and implicit bailout guarantees (e.g., AIG). Households lack the ability to properly monitor the risks and long-term solvency of insurers, but depend on insurance products to smooth the idiosyncratic and aggregate risks they face. Fragility of insurers could affect households' propensity to buy insurance, which could result in increases in precautionary savings and welfare loss for households. Life insurers are also among the largest investors in many fixed income markets and a source of long-term capital for corporations, municipalities, and other agencies. Fragility of insurers could give rise to fire sale externalities and increases in borrowing costs, which could have contractionary consequences for the economy.

While the rationale for having regulation is clear, developing appropriate standards to adequately measure the risks of financial products is challenging. In particular, there are several reasons the regulatory framework governing VAs could have inconsistencies. VAs are relatively new products, gaining prominence only since the 1990s. VAs also have more complex payoffs and are hedged using different instruments relative to traditional insurance products. My setting shows that regulators, when faced with the full complexity of financial products, could have a difficult time getting the treatment of the risks exactly right, and more so across products, risk factors, and under different market conditions.⁴⁴

More broadly, the types of regulatory incentives I identify are prevalent in other parts of the financial sector. For example, the regulatory framework governing insurance companies in Europe, known as Solvency II, has important deviations from market values. Differences also exist in the risk sensitivities and the treatment of derivatives in the model-based versus the standardized frameworks under Basel III. The findings of my paper could, therefore, extend to banks and insurers globally.

References

- Acharya, V., and M. Richardson. 2009. Causes of the financial crisis. *Critical Review* 21:195–210.
- Acharya, V., L. H. Pedersen, T. Philippon, and M. Richardson. 2017. Measuring systemic risk. *Review of Financial Studies* 30:2–47.
- Adrian, T., and H. S. Shin. 2014. Procyclical leverage and value-at-risk. *Review of Financial Studies* 27:373–403.
- Becker, B., and V. Ivashina. 2015. Reaching for yield in the bond market. *Journal of Finance* 70:1863–902.
- Becker, B., M. M. Opp, and F. Saidi. 2021. Regulatory forbearance in the U.S. insurance industry: The effects of removing capital requirements for an asset class. *Review of Financial Studies*. Advance Access published September 3, 2021, 10.1093/rfs/hhab102.
- Begenau, J., M. Piazzesi, and M. Schneider. 2015. Banks' risk exposures. Working Paper, Stanford University.
- Black, F. 1976. The pricing of commodity contracts. *Journal of Financial Economics* 3:167–79.
- Black, F., and M. Scholes. 1973. The pricing of options and corporate liabilities. *Journal of Political Economy* 81:637–54.
- Blake, D., A. G. Rossi, A. Timmermann, I. Tonks, and R. Wermers., 2013. Decentralized investment management: Evidence from the pension fund industry. *Journal of Finance* 68:1133–78.
- Bohn, J. G., and B. J. Hall. 1999. The moral hazard of insuring the insurers. In *The financing of catastrophe risk*, eds. K. A. Froot, 363–90. Chicago: University of Chicago Press.
- Breeden, D., and S. Viswanathan. 1990. Why do firms hedge? An asymmetric information approach. *Journal of Fixed Income* 25:7–25.
- Bretscher, L., L. Schmid, I. Sen, and V. Sharma. 2020. Institutional corporate bond demand. Research Paper, Swiss Finance Institute.

⁴⁴ There have been growing efforts to identify the weaknesses in the statutory framework and steps required to properly align the regulatory risk sensitivities with economic exposures (Oliver Wyman 2018).

- Cochrane J. H., and M. Piazzesi. 2005. Bond risk premia. *American Economic Review* 95:138–60.
- Coval, J., J. Jurek, and E. Stafford. 2009. The economics of structured finance. *Journal of Economic Perspectives* 23:3–25.
- Cox, J. C., J. E. Ingersoll, and S. A. Ross. 1985. A theory of the term structure of interest rates. *Econometrica* 53:385–407.
- Credit Suisse. 2012. US variable annuities, Credit Suisse Connections Series.
- DeMarzo, P. M., and D. Duffie. 1992. Corporate incentives for hedging and hedge accounting. *Review of Financial Studies* 8:743–71.
- Domanski, H. S. Shin, and D., V. Sushko. 2017. The hunt for duration: Not waving but drowning? *IMF Economic Review* 65:113–53.
- Drechsler, I., A. Savov, and P. Schnabl. Forthcoming. Banking on deposits: Maturity transformation without interest rate risk. *Journal of Finance*.
- Ellul, A., C. Jotikasthira, and C. T. Lundblad. 2011. Regulatory pressure and fire sales in the corporate bond market. *Journal of Financial Economics* 101:596–620.
- Ellul, A., C. Jotikasthira, C. T. Lundblad, and Y. Wang. 2015. Is historical cost a panacea? Market stress, incentive distortions and gains trading. *Journal of Finance* 70:2489–538.
- Ellul, A., C. Jotikasthira, A. V. Kartasheva, C. T. Lundblad, and W. Wagner. 2020. Insurers as asset managers and systemic risk. Research Paper, Kelley School of Business.
- Froot, K. A., and P. G. O’Connell. 1999. The pricing of US catastrophe reinsurance. In *The financing of catastrophe risk*, chap. 5, 195–232. Chicago: University of Chicago Press.
- Froot, K. A., D. S. Scharfstein, and J. C. Stein. 1993. Risk management: Coordinating corporate investment and financing policies. *Journal of Finance* 48:1629–58.
- Froot, K. A., and J. C. Stein. 1998. Risk management, capital budgeting and capital structure policy for financial institutions: An integrated approach. *Journal of Financial Economics* 47:55–82.
- Ge, S. 2022. How do financial constraints affect product pricing? Evidence from weather and life insurance premiums. *Journal of Finance* 77:449–503.
- Ge, S., and M. S. Weisbach. 2021. The role of financial conditions in portfolio choices: The case of insurers. *Journal of Financial Economics* 142:803–30.
- Hanson, S. G. 2014. Mortgage convexity. *Journal of Financial Economics* 113:270–99.
- Hoffmann, P., S. Langeld, F. Pierobon, and G. Vuilleme. 2019. Who bears interest rate risk. *Review of Financial Studies* 32:2921–54.
- Junus, N., and Z. Motiwalla. 2009. A discussion of actuarial guideline 43 for variable annuities. Technical Report, Milliman.
- . 2010. Implications of actuarial guideline 43 for variable annuity product design and risk management. Technical Report, Milliman.
- Kirti, D. 2017. When gambling for resurrection is too risky. Working Paper, IMF.
- Klingler, S., and S. Sundaresan. 2019. An explanation of negative swap spreads: demand for duration from underfunded pension plans. *Journal of Finance* 74:675–710.
- Koijen, R. S. J., and M. Yogo. 2015a. The cost of financial frictions for life insurers. *American Economic Review* 105:445–75.
- . 2015b. Risks of life Insurers: Recent trends and transmission mechanisms. In *The economics, regulation, and systemic risk of insurance markets*, chap. 4. Oxford, UK: Oxford University Press.
- . 2016. Shadow insurance. *Econometrica* 84:1265–87.
- . 2022. The fragility of market risk insurance. *Journal of Finance* 77:815–62.

- Lawsky, B. M. 2013. Shining a light on shadow insurance: A little-known loophole that puts insurance policyholders and taxpayers at greater risk. New York State Department of Financial Services.
- Lee, S., D. Mayers, and C. W. Smith, Jr. 1997. Guaranty funds and risk-taking evidence from the insurance industry. *Journal of Financial Economics* 44:3–24.
- Litterman R. B., and J. Scheinkman. 1991. Common factors affecting bond returns. *Journal of Fixed Income* 1:54–61.
- Milliman. 2021. Telephone interview with Milliman, August 4, 2021.
- Miranda-Agrippino, S., and H. Rey. 2020. US monetary policy and the global financial cycle. *Review of Economic Studies* 87:2754–76.
- Nance, D. R., C. W. Smith, and C. W. Smithson. 1993. On the determinants of corporate hedging. *Journal of Finance* 48:267–84.
- Office of Financial Research (OFR). 2016. Mind the gaps: What do new disclosures tell us about life insurers' use of off-balance-sheet captives?
- Oliver Wyman. 2018. Variable annuity statutory reserve and capital reform.
- Perli, R., and B. P. Sack. 2003. Does mortgage hedging amplify movements in long-term interest rates? Finance and Economics Discussion Series 49, Board of Governors of the Federal Reserve System (U.S.).
- Purnanandam, A. 2007. Interest rate derivatives at commercial banks: An empirical investigation. *Journal of Monetary Economics* 54:6.
- Rampini, A. A., and S. Viswanathan. 2010. Collateral, risk management, and the distribution of debt capacity. *Journal of Finance* 65:2293–322.
- Rampini, A. A., S. Viswanathan, and G. Vuillemey. 2020. Risk management in financial institutions. *Journal of Finance* 75:591–637.
- Reinhart, V., and B. P. Sack. 2000. The economic consequences of disappearing government debt. *Brookings Papers on Economic Activity* 31:163–220.
- Sen, I., and D. Humphry. 2018. Capital regulation and product market outcomes. Working Paper, Bank of England.
- Sen, I., and V. Sharma. 2020. Internal models, make believe prices, and bond market cornering. Working Paper, Harvard Business School.
- Smith, C. W., C. W. Smithson, and S. D. Wilford. 1990. *Managing financial risk*. New York: Harper and Row.
- Smith, C. W., and R. M. Stulz. 1985. The determinants of firms' hedging policies. *Journal of Financial and Quantitative Analysis* 20:391–405.
- Stulz, R. M. 1984. Optimal hedging policies. *Journal of Financial and Quantitative Analysis* 19:127–40.
- . 1990. Managerial discretion and optimal financing policies. *Journal of Financial Economics* 26:3–27.
- Tiller, J. E., and D. F. Tiller. 2009. *Life, health and annuity reinsurance*. Winsted, CT: ACTEX Publications.
- Vuillemey, G. 2019. Bank interest rate risk management. *Management Science* 65:5933–56.